

Cambridge International AS Level
Information Technology 9626

Paper 1 Theory

High-signal mark schemes

Mark schemes for the 110 questions in "AS Level IT 9626 - Paper 1 high-signal questions.pdf", numbered the same way (1.1, 1.2, ...) and labelled with the paper each question came from.

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1 Data processing and information

1.1 9626/12 Feb/March 2025 · Question 1

Question	Answer	Marks
1	One mark per bullet point to a maximum of five marks. - Information is data with context (1) and meaning (1) - This could be about e.g. a student/lecturer/employee/payroll system (1) Any examples of the data presented with context or meaning (interpretation) (max 5 – each item of data may only be used once) e.g. - Chandra could be the first name. (1) 59 could be their age. (1)	5

1.2 9626/12 May/June 2024 · Question 7

Question	Answer	Marks
7	Eight from: Simple statement of the difference between the two (1): e.g. - Direct data is a primary source, indirect data is a secondary source. - Description of how primary and secondary are collected. Advantages MAX SIX - Direct data source/direct data is reliable/trusted (1st) – because we collected it (1) - Data needs to represent the views of a specific target audience (accept other terminology) (1st) – we can make sure a cross-representation of the group is sampled/included in the sample (1) – and so avoid sample bias (1) ONLY Specific/relevant data is collected (1) - Indirect data includes irrelevant data (do not allow opposite) (1) - Person/people collecting the data only need to collect as much/or as little (quantity) as is needed//to suit your needs (1st) – whereas with indirect data sources, the original purpose may have not collected enough//collected too much data (1) Collected data could be sold on to other organisations (1st) – thereby helping to recoup some of the expenses/money spent (1) - Direct data is (more) up-to-date (1)	8

Disadvantages MAX SIX

- The sample size may be small (1st)
 - whereas indirect data sources tend to be larger sets of data (1)
- Some people/groups may not be accessible by the person collecting the data (1st)
 - whereas with indirect data sources, would allow data from these groups to be collected/gathered (1)
- With direct data collection, people may not be at home/available to be interviewed (1st)
 - can result in a smaller sample size (1)
- Original data may not be possible to collect because of the time of year (1)
- Indirect data allows historic data to be used (1)
- Direct data can take a long time **to collect** (1)
- With direct data, data needs to be collated (NOT “processed”) (1)
- Collecting **direct** data involves the expense of travelling/photocopying/printing questionnaires//any expense to do with a direct data method (1)

1.3 9626/12 May/June 2026 · Question 1

Question	Answer	Marks	Guidance
1(a)(i)	ONE from: - Questionnaire (e.g.,) where questions are asked//customers are asked their opinions (1) - Interview is (e.g.) a face to face meeting/questions are asked (1) - Observation where situations are watched (1) - Surveys (e.g.) where questions are asked (1)	1	Marks may only be awarded where the method has been stated. Answer MUST do more than identify a method
1(a)(ii)	ONE from: e.g. - Textbooks (1) - Journals (1) - Websites//research published online (1) - Someone else's (e.g.) questionnaire (1) - Market research DONE FOR someone else (1) - Government statistics (1) - Newspapers/magazines (1)	1	Do Not Award: - Weather data - Census data - Electoral register - Any source that would not include expenditure patterns - Any source where it is not clearly someone else's
Question	Answer	Marks	Guidance
1(b)	Four marks available: TWO from relevance: - Reduces the chance of excess/unnecessary data (accept converse) (1) - The businesswoman has control over what information is gathered (accept exemplification) e.g. ask questions about sports goods (1) - The businesswoman can design the method/stated method, so can ensure that the (e.g.) questions asked gather the information she needs (1) - Can ensure data is gathered from the town where she will open the shop (1) - Data is obtained/collected with the set purpose in mind (1)	4	

	TWO from age: • The questions are being asked now (1) • The age of the data is a measure of how long ago it was collected (1) • Use of a direct data source means that data can be/is more up-to-date (1) • Patterns of behaviour/attitudes change over time//what was correct (e.g.) twenty years ago is not necessarily correct now (accept converse about direct data) (NOT 'things could have changed' – TV (1)		
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Question	Answer	Marks	Guidance
1(c)	Four marks available. Two per matched explanation: Identify reason (1) and state why it is a reason/why beneficial (1) Areas for consideration (Note: may also be used as expansion. Mark as candidate's intension) • Larger sample size (must be the concept of more data/more people asked) • Less biased/more accurate/more reliable NB accuracy or reliability could be used as an extension of (e.g.) less biased for two marks • Wider variation/type of people • Seasonal nature of some sports • Convenience • No time to collect data so use someone else's • Avoidance of a cost (1) such as (1) • Quicker to achieve results (of analysis) Exemplar answers • The sample size may be larger (1) therefore the results gathered are less likely to be biased (1) • She may not have the time to collect the data (1) and so uses data gathered by someone else (1)	4	

Question	Answer	Marks
8(a)	Four from: - When a message/information is intercepted//accessed it is unreadable/ can't be understood ... (1) ... so, therefore, it is useless (1) ... to a hacker (accept alternative examples of 3rd party or just 'third party') (1) - It protects customers when they bank/shop online/any other suitable example from any area of use (MUST BE protecting data) ... (1) ... as the data cannot be used for identity theft (1) - Data can only be decrypted by the receiving computer if it has the private/decryption key (1) Max. three marks if bullets/list of points	4

Question	Answer	Marks
8(b)	Up to FOUR marks available: Advantages (MAX 3) - It is a faster process//quicker to encrypt compared to asymmetric (1) as it has less mathematical complexity (1) - It requires less computational power than asymmetric (1) Disadvantages (MAX 3) - Does not have a different key for encryption and decryption//sender and receiver use the same key (1) therefore only one key needs to be stolen (1) ... making it less secure (1) - The encryption key cannot be generally published (1) - The encryption key must be sent to the receiver of the message (1) - Key could be intercepted/stolen (by a hacker) whilst being sent ... (1) ... who can then decrypt any message encrypted by the sender (accept any reasonable implication) (1) Must have at least one of each to gain full marks Max. three marks if bullets/list of points	4

1.5 9626/13 Oct/Nov 2024 · Question 5

Question	Answer	Marks
5	FOUR from: • The public key is used to encrypt the data (1) • The private key is used to decrypt the data (1) • The public key is published to everyone (1) • The private key is sent separately to the receiver (1) • The private key is kept secret (1) • Anyone with a copy of the public key can encrypt information (1) • Only the private key holder can decrypt the information (1) • It is not possible to decrypt the data from the public key (1)	4

1.6 9626/12 May/June 2024 · Question 4

Question	Answer	Marks
4	Eight from: Data answers • Transmitted data/file is encrypted (1) so that a third party cannot understand the data//data/file is useless to a hacker (1) • Only the user's computer and the secure/receiving server can recognise the data/file (1) • SSL/TLS use two keys//private and public keys (1) Certificate/process answers (may be answered in any order) • Websites often use (SSL/TLS) certificates to verify their authenticity (1) • Web server is asked to identify itself (1) • The web server sends a copy of the (SSL/TLS) Certificate (1) • Certificate Authorities are used to ensure the certificate is real/authentic of the certificate/provides a list of approved certificates (1) • The browser checks to see if the SSL/TLS Certificate is trusted (1) • If the SSL/TLS Certificate is trusted, then the browser sends a message to the web server (1) • If the SSL/TLS Certificate is not trusted, the browser will refuse to open the website (1) and will display a message to the user (1) • (IF TRUSTED) The server then responds to the browser with a digitally signed acknowledgement (1) to start an SSL/TLS encrypted session (1) • Session keys are set up (1) • SSL/TLS requires a handshake to be carried out/the browser and the website carry out a (SSL/TLS) handshake. (1)	8

1.7 9626/13 Oct/Nov 2024 · Question 4

Question	Answer	Marks
4(a)	THREE from: - Verification is checking that data that has been/is being entered into a computer has been copied correctly (1). - Validation is checking that the data entered is reasonable/sensible/valid/follows a set of rules. (1) - Verification will not pick up the fact that the original data might be incorrect. (1) - Validation can not tell if the data has been entered incorrectly (1) - Verification would pick up a transposition error. (1) - Verification would pick up a transcription error. (1)	3
4(b)(i)	TWO from: - It is a validation method (1) - It checks that data is possible/sensible for the context (1) - It checks that data is between an upper and lower acceptable value (1) - Sensible example (1)	2
4(b)(ii)	TWO from: - It is a verification method (1) - It is used to avoid/pick up transcription errors (1) - Two versions of the data are entered and compared (1st) - If a discrepancy between the two versions is found, user will be alerted (1) - Sensible example (1)	2

1.8 9626/13 Oct/Nov 2023 · Question 4

Question	Answer	Marks
4(a)	Three matched pairs from: - Might enter an account number with 9 or 11 digits/less than/more than 10 (1st) Length check to make sure that the account number is exactly 10 characters (1) - Might enter a wrong digit (1st) - Use a check digit which creates a new digit by performing a calculation on individual digits of the account number and appending the resulting digit (1) - Might omit one of the account numbers (1st) Presence check would make sure account numbers are entered. (1)	6
4(b)	Three from:	4

	• A key number would be calculated (1) • All the account numbers/numbers in the account number would be added together (1st) • To create a single number (1) • The data/file would be transmitted along with the hash total/key number (1) • The hash total is recalculated once received (1) • The original hash total is compared with the new one (1) • If they do not match an error has been made (1) One from: • A control total checks fields that are used for calculations whereas a hash total is used for fields that hold information (1) • The field holds data, and will not be used for calculations (1)	
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1.9 9626/11 Oct/Nov 2023 · Question 11

Question	Answer	Marks
11	Eight marks available: Advantages max 6 • Orders are/can be processed straight away (1st) • so process of filling/completing orders can start sooner (than for batch processing) (1) • With online processing errors are revealed immediately (1) • With batch processing systems if there is an error it is only revealed when processing takes place (1st) • Delay in finding the error (1) • If the database needs to be interrogated details can be retrieved immediately (1st) • Any example of why immediate interrogation would be required – e.g a sales invoice/order may be required when a customer queries a bill (1) • With online systems they are available whenever necessary (accept converse argument) (1) • Data is accurate all the time (1) Disadvantages max 6 • Extra hardware requirements/input devices/workstations of an online processing system can make it more expensive (than batch processing) (must clarify at least one process for the comparison) (1) • Online processing has to be active for the whole day (1) • There must be workers available (to enter data) at all times (1st) • more employees are needed (1) • company wage bills can be higher (1) • Batch processing systems only require a limited number of employees to enter all the data (1) • It is more expensive than batch processing as it uses much more computer processing time to update files (1) • With batch processing, processing occurs at a time convenient to the employees responsible for entering the data (1) • Batch processing only uses computers when the demand on the system is low (1st) • therefore entering data does not slow the network down (1) Max. six marks if bullets/list of points	8

Question	Answer	Marks
3	Seven from: Sale • Computer reads the barcode number (1) • Computer searches for this barcode number in the product file/database (1) • If this barcode number is found, then number in stock is reduced (1) Order • Computer compares the number in stock to the re-order level/set value (1) • If the number in stock is not equal (or lower) to the re-order level/set value another barcode is input//systems returns to the start (1) • If the number in stock is equal/lower than the re-order level//a threshold/set value (1st) – the computer creates an (automatic) order (1) • Item is flagged to be reordered (1) • Computer looks up/finds the re-order quantity (for that product) (1) • Computer looks up/finds the supplier (reference) number (for that product) (1) • Computer searches/checks (the supplier file) for the record of the supplier (1) • Computer orders automatically (1st) – using the supplier's contact details. (1)	7

2 Hardware and software

2.1 9626/12 Feb/March 2026 · Question 4

Question	Answer	Marks
4	SIX marks available: ONE mark per identified characteristic, ONE mark for the difference: • Longevity (1st) Mainframe computers last longer than supercomputers (1) • RAS/Reliability, availability and serviceability (1st) Mainframe computers are more reliable/available/serviceable than supercomputers (1) The term RAS is NOT used to assess supercomputers but does refer to mainframe computers (1) • Security (1st) Mainframe computers are more secure than supercomputers (1) • Performance metrics//any statement of processing (1st) Supercomputers are measured in FLOPS/TEPS and mainframes are measured in MIPS (1) Supercomputers are more powerful computers than mainframe computers (1) • Volume of input, output and throughput (1st) A mainframe has a higher rate of throughput//can deal with a large number of simultaneous transactions (1) • Fault tolerance (1st) Mainframe computers have a high level of fault tolerance (1) Mainframe computers allow hot swapping (1) Mainframes have a high mean time between failures (MTBF) (1) • Operating system (1st) Super computers tend to only use Linux (derivative) operating systems (1) Mainframe computers can have multiple operating systems (1) • Number of processors (1st) Mainframe computers have multiple processors (1) • Cores (1st) Supercomputers have more cores than mainframe computers (1)	6

Question	Answer	Marks
3(a)	Three from: - Whether a computer will repair itself if an error occurs (1) - Whether a computer can continue to operate even if one (or more) of its components has failed (1st) - without affecting the operation of the system/without downtime (award once only) (1) - Whether a computer's operating quality can reduce, but does not fail completely (1) - Whether when a hardware component/processor fails, another takes over/elements are hot swappable (1st) - without affecting the operation of the system/without downtime (award once only) (1) - Whether computers have two different instances/installations/versions of software (1st) - if there is a problem with the first instance/installation/version, the other version is automatically run (1) - this (characteristic) does not apply to mainframes (1)	3
3(b)	Two from: - Computers generate heat/become overheated (1). - When a computer overheats, it will address this issue (1). - Overheating of mainframes and supercomputers is a problem (1st) - due to the quality and quantity of processors generating a lot of heat (1) - hence the need for cooling systems (1) - Air cooling systems have been used but uses up a lot of power/are very expensive to build and operate (1) - Water cooling systems are more cost-effective/more efficient (1) - in removing any excess heat (1) - Many data centres are now being built in cooler parts of the world (1)	2
3(c)	Three from: - The amount of data that has to be processed is very large (1st) as countries tend to have a lot of people (1) - The amount of data is increasing (1st) as each census produces more data (to add to what already exists)/world populations are rising (1) - All data needs to be processed in a short period of time (1st) - and mainframes have a high speed of processing data (1) (As more data is added), more powerful machines are needed (1st) - and mainframes are easily scalable (1) - Mainframes are very reliable (1st) MAX two marks if answers are all only first marks.	3

Question	Answer	Marks	Guidance
5	SIX from: MAX 5 if no reference to weather forecasting Benefits (Max four) • Very fast processing//has a high FLOPS (1) • Weather forecast can be produced quickly (1) • Can handle a great deal of information (NOT store) (1) • Produce more/very detailed forecasts (1) • Consider a wide range of possible scenarios (1) • Decision is data driven (1) Drawbacks (Max four) • Supercomputers struggle to predict extreme events//working with extreme data (1) • Lacks human intuition (1) • Supercomputers have a limited lifespan (1) • Supercomputers are expensive to buy (1) • Supercomputers require specialist maintenance teams (1) • Supercomputers are very large//take up a lot of space (1) • Supercomputers require a lot of heat management//generates a lot of heat (1)	6	Do Not Accept: • ...and supercomputers can do this Some marks are awarded based on the use of <u>superlatives</u>. Candidates MUST have a clear awareness of the absolute vastness of the processing speed, or of the amount of information able to be handled. Therefore, if this is established, words such as 'a lot' or 'a great deal' can be awarded.

Question	Answer	Marks
5	Eight from: <i>Advantages</i> Once it is compiled a program does not have to be translated again A compiled program can run faster than interpreted software Compiling a program is a much quicker process /translates it faster than interpreting... ... as whole program is translated all in one go /all at onceunlike interpreter translates one line at a time Once the program is already compiled into machine language, there is no translation software that the user has to keep up to date Compilers also improve security for programmers... ...makes it much harder to copy code/machine code is difficult to understand and alter... ...so fewer risks of copyright infringement... ... with an interpreted program, the source code can always be seen so greater risk of software copyright infringement <i>Disadvantages</i> Can only be used on one operating system ... source code can be interpreted on any operating system provided interpreter is present... ...will need to use an interpreter if program is to be shared... It is slow/(more) difficult to debug the program/find errors... ...the whole program needs to be compiled again after an error is corrected... ...error messages are output at the end... ...whereas with an interpreter error messages are output <u>as they are found</u> It uses up more memory than an interpreter... ...as whole program has to be loaded... ...interpreter only requires a few statements of the program to be in memory Unlike an interpreter, does not allow small pieces of code to be tested to make sure they work before continuing with the rest of the program More likely to crash the computer as it is running directly on the CPU Must have at least two of each to gain full marks Must be a proper discussion to obtain full marks Max. six marks if bullets/list of points Must have expansions or comparisons to be a proper discussion	8

2.5 9626/13 May/June 2025 · Question 8

Question	Answer	Marks
8(a)	One mark per bullet point to a maximum of four marks. - It operates/controls a device attached to the computer (1) - It provides an interface between operating systems/applications software and the device (1st) - Which handles the translation of requests between a device and the operating system (1) - It allows an operating system to detect/identify the peripheral device (1) - Activates the device when it is needed and deactivates it when it is not needed (1) - Logs and reports errors (1) - Defines where data must be cached (1)	4

Question	Answer	Marks
8(b)	One mark per bullet point to a maximum of four marks. - Translates a high-level language into machine code (1) - Creates an executable file (1) - Identifies errors in code (1) - Stops interpreting code when an error is found//shows errors as it translates (1) - Errors are dealt with as they are found//immediately (1) - Uses less RAM as only one line at a time is stored/processed (1)	4

2.6 9626/11 Oct/Nov 2024 · Question 1

Question	Answer	Marks
1(a)	THREE marks available Max TWO marks: - To prepare a data storage medium for initial use (1) - The purpose of low-level formatting is to prepare the disk surface to receive data/allow the user to save data/files (1) - High-level formatting creates a file system on the disk (1) - Allows the operating system to use the disk space to store and access files (1). Max TWO marks: - Low-level formatting divides the disk surface into tracks/sectors/cylinders (1) - Write heads magnetise the disk areas (1) - Tracks are numbered starting from 0 (1) - A gap is left between tracks (1) - Each track is organised into numbered sectors, starting at 1 and separated by gaps (1)	3
1(b)	THREE marks available:	3

	- Identifies Lossless and identifies Lossy compression (1) Max TWO marks - Reduces the overall number of bits/bytes (in a file)//reduces file size (1) - To increase available storage space on a hard drive (1) - File compression programs remove redundancy (1) Max TWO Marks - A lossless file compression program uses codes to represent repeated Information (1) - Lossless compression removes meta data (1)	
1(c)	THREE marks available: Max TWO marks: - It speeds up the access time to a file (1). - To create larger regions of (contiguous) free space (1) Max TWO marks: - Organises the contents of the disk into the smallest number of contiguous regions/fragments (1) - The data blocks of a file/defragmented files are moved around to bring whole/all the parts of a file together (1)	3

2.7 9626/13 Oct/Nov 2022 · Question 3

Question	Answer	Marks
3(a)	Three from: Scans the computer to make sure it isn't infected with a virus/to find viruses (1) - Can scan whole disk or individual files (1) - Can be set up to scan automatically//run at a set time (1) - Compares with a database/existing viruses (1) - Detects changes in behaviour of files (1) - Quarantines the infected programs (1) - Reports results of scan to user (1) - It will ask the user whether or not they want to delete the infected programs (1) - Does background scans of downloads and attachments (1) Automatically update itself (1)	3
3(b)	Three from:	3

	- Creates additional copies of files/data/databases/hard disks/ network servers (accept 'duplicates') (1) - Backed up//copy of (e.g.) files are still available in the event of data loss//if computer is corrupted/computer fails/(accidental) deletion/other example of data loss (1) - Can use these copies to restore the original contents in the event of data loss//if computer is corrupted/computer fails/(accidental) deletion/other example of data loss (1) - Asks user to enter type of back-up (1) - Can happen automatically (accept reverse argument) (1) - Asks if you wish to restore the back-up (1) - Asks if you wish to verify the back-up (1) - Asks when back-up is to take place/frequency of backups (1) - Ask which medium the user wishes to store/restore backup (1)	
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2.8 9626/12 Feb/March 2026 · Question 9

Question	Answer	Marks
9	EIGHT marks available: Areas for consideration (one mark each) - Ability to dictate how the product is tested//Extent to which testing has occurred//Potential for existence of bugs (1) - Extent to which user is/can be involved in testing (1) - Extent to which software has features as required//meets the requirements//tailored for the business (1) - Extent to which software includes unnecessary features//ONLY required features (1) - Compatibility with current system (1) - Copyright ownership//ability to sell on (1) - Who pays/paid development costs (1) - Whether immediately available for use or not (1) - Availability of support (online forums etc.) (1) - Updates (costs of or availability of) (1) - Reliance on continued existence of vendor going forward//product ceases to be available (1) - Scalability (1) Marks may only be awarded where the point made is correctly linked to the type of software. MAX six marks where only custom-written or off-the-shelf are considered.	8

Question	Answer	Marks
9	Eight from: Advantages of GBI/disadvantages of GUI • Can be advantageous to disabled users ... (1) ... who maybe cannot control their hands accurately//have limited mobility (1) ... could make gestures using their limbs/eyes (1) • For reasons of hygiene a doctor/food worker may not be allowed to touch a display or device ... (1) ... gestures control the device/ allow them to control devices without touching anything (1) • Gestures may be a quicker way of initiating a response from a device (1) • Some gestures may be unintentional but still initiate a reaction from the device (1) • Gesture based interface does not require person to be sat in front of computer/device so can operate computer/device from a distance (1) • Enhanced gaming experience (1) Disadvantages of GBI/advantages of GUI • GUIs tend to be more accurate than gestures (1) • Certain gestures may not be socially acceptable/user could feel uncomfortable making gestures//reduced privacy (1) • Some gestures may be unintentional but still initiate a reaction from the device//GUIs require actions to be deliberate (1) • Gesture based is less effective when several users/background movement involved (1) • In certain lighting conditions gesture-based interface may misinterpret/fail to register a gesture (1) • GUIs can be used in any lighting/dark situations (1) • Increased danger of injury (1) Max. six marks if bullets/list of points	8

Question	Answer	Marks	Guidance
9(a)	SIX marks available: Effect of icons • Avoids the need to read (1) • Graphical icons engage on an emotional level//are more appealing//visually appealing (1st) Therefore, more likely to be completed (1) Results are more reliable/accurate (1) • Less time is required to record customer satisfaction (1st) Therefore more likely to respond (1) Results are more reliable/accurate (1) • Graphical icons are simple to understand (1st) Therefore people will not need to queue/stop to be able to read them (1) Results are more reliable/accurate (1) • Graphical icons are universal/international/not language dependant (1st) Therefore more likely to respond (1) Results are more reliable/accurate (1) • Icons will be more widely understood (1st) Therefore more likely to respond (1) Results are more reliable/accurate (1) Languages • No need to use a 'select language button' (1st) Therefore people will not need to queue/stop to be able to read them (1) Therefore more likely to respond (1) • No guarantee that everyone speaks the same language//the language of the country in which the airport is (1) • In an airport, there will be a range of languages spoken (1)	6	Only award answers that support the use of a GUI as a method Do not award • repetition of explanations • answers about the system finding icons easier to process//.impact of use of icons on processing etc. • Users of the airport may be in a rush • GUI more easy to understand • Easier to navigate/use

Question	Answer	Marks	Guidance
9(b)	FOUR marks available: • Lots of noise (may interfere with vocal/vocal may not be clear) (1st) Therefore the computer may misinterpret what has been said (1) • Can not be used by certain disabilities (1) because these users cannot (e.g.) speak (1) • Would need to support all/many languages//may not support a specific languages (1st) Would need a very large database of languages/dialects (1)	4	Mark first two answers only. Do Not Accept: • Will have difficulty with accent

3 Monitoring and control

3.1 9626/12 May/June 2022 · Question 6

Question	Answer	Marks
6(a)	Humidity sensor combination of moisture sensor and temperature sensor/measures how much moisture is in atmosphere together with temperature – 1 mark Used in weather stations to measure relative/absolute humidity /air humidity /any suitable example – 1 mark Sound sensor converts sound waves into voltages/electrical signals – 1 mark Used in environment monitoring systems to measure noise pollution /any suitable example – 1 mark	4
6(b)	Two from: The accuracy of all sensors reduces after a period of time... ...due to constant use in/exposure to the atmosphere/liquids that are being used Regular calibration helps to maintain the accuracy of the sensors/helps to make sure sensors produce accurate/correct/reliable readings Other devices within a monitoring system may deteriorate over time resulting in a need for recalibration	2
6(c)	Five from: Only one measurement point/reading with one point calibration whereas with two point calibration at least two readings are taken One point calibration is appropriate for sensors used to measure a value that is constant/never changes... ...whereas two-point calibration is used with sensors that measure a constantly changing variable With one point calibration the offset only is calculated/ the sensor reading is subtracted from the known value only... ...the offset is then added to every subsequent reading With two point calibration sensitivity needs to be included not just an offset Two point calibration is used where there is a linear relationship between the two readings The slope of the line connecting the two readings is compared with the slope of the line of the pre-calibrated sensor/the range values of the sensor is compared with the range values for the standard Two point calibration compensates for both offset errors and sensitivity/slope errors	5

3.2 9626/11 and 9626/13 Oct/Nov 2025 · Question 5

Question	Answer	Marks
5(a)	Two from: • Sensor accuracy changes with time/use because of wear and tear from environmental circumstances/constant use (1) • Sensors might not be adapted to the required environment because the conditions can change (1) • Calibration maintains the accuracy of the sensors to ensure they produce reliable/accurate/correct readings (1) • New sensors need to be setup/calibrated to ensure they meet the appropriate requirements because they will be set to factory settings//have preset parameters (1) • Other components that are part of the system/impact on the reading may affect the reading so making it inaccurate (1)	2
5(b)	Five from: e.g. • Only one measurement point/reading with one-point calibration//with multipoint calibration at least three readings are taken (1) • One-point calibration is used with sensors that measure a single value// multipoint calibration is used with sensors that measure a range of values that have a non-linear relationship (1) • One-point calibration requires a simple calculation to take account of the error//multipoint calibration requires a complex calculation (1) • They both mitigate/reduce impact of structural errors in readings//inaccurate readings (1) • They both need to be done regularly (1) • They both compare values with known/correct/accurate measurements (1)	5

3.3 9626/11 May/June 2025 · Question 7

Question	Answer	Marks
7(a)	One mark per bullet point to a maximum of six marks. • A moisture/humidity sensor/tensiometer (1st) – is used to measure absolute and relative humidity (1) – measures the amount of moisture in the air (1) • A wind sensor/anemometer (1st) – is used to measure the direction of the wind (1) – used to measure strength of wind (1) • A raindrop/raingauge/rain sensor (1st) – is used to measure rainfall (1) • A light sensor/photodetector (1st) – is used to measure/sense light (1) • A pressure/barometric sensor (1st) – is used to measure pressure values and height changes. (1) • A temperature/NTC thermistor/RTD thermocouple/IC sensor (1st) – is used to measure temperature (1)	6
7(b)	Six marks from:	6

	• The data will be sent to the computer for processing (1) • Via Analogue Digital Convertor (ADC) (1) • ADC will convert the analogue data to digital (1st) So, the computer can read it (1) • It can create graphs/charts (1st) – To show monthly trends (1) • It can do calculations (1st) – Such as averages/max min (1) • It can produce predictions (1st) – Based on historical of weather data (1) • It can produce reports of weather trends (1st) – Such as global warming (1) • It can produce weather warnings (1st) – From its collected data. (1)	
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3.4 9626/13 May/June 2024 · Question 6

Question	Answer	Marks
6(a)	Six from: • Sensor used to gather data (1) • Identifies ONE suitable sensor for collecting information on presence OR absence of pollution (1) • Data collected is analogue (any awareness) (1) • Analogue to Digital Convertor converts data to digital (1st) • So can be used/understood by the computer (1) • Computer analyses/processes data (1) • Results presenting in appropriate format/tables/graphs/charts (1)	6

Question	Answer	Marks
6(b)	Eight from: <i>Advantages MAX SIX:</i> • Data is continually monitored//no reason for monitoring not to occur (may exemplify) (1st) • if an event happens it should not be missed (1) • Computer systems can monitor multiple sensors simultaneously (1) • The logged data is already in electronic form (1st) • no transcription errors/no human error (1) • Does not need to be entered/typed in (1st) • which saves time (1) • Personnel can do other useful tasks (1) • Data is immediately sent to the computer (1st) • The data can be displayed immediately/in real time (1) • The data can be analysed immediately/in real time (1) • Can monitor 24/7 (1) • No risk to humans/equivalent answer (1) • Can be monitored remotely (1)	8

	Disadvantages MAX SIX: - Any interruptions to the power supply could cause sensor to stop working//collection of data to be missed/not happen (1) - Any damage/failure of the equipment could cause collection of data to be missed/not happen/be inaccurate (1) - The maintenance of equipment can be expensive/dangerous/time consuming//requires expertise/training. (1) - Data logger can not react to emergencies//unusual circumstances (1)	
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3.5 9626/11 Oct/Nov 2023 · Question 6

Question	Answer	Marks
6(a)	Two from: - An actuator provides the output from a control system (1) - Is the mechanical part of a control system (1) - An actuator controls/operates a device (1st) - such as opening a valve/activating a motor/switch (1) - It turns an electrical signal (from a microprocessor) into movement/action/physical (1)	2
6(b)	Two marks from: - Measures the change in a magnetic field(1st) - caused by the presence of a metallic/ferromagnetic object/body of the vehicle (1) OR - Identifies a magnetic field from a car (1st) - caused by the car being metal (1) Generic answers: - Are used (at the entrances to car parks) to control barriers (1) - Are used to detect the number/position of spaces available//whether a (parking) space is occupied (1)	2
6(c)	Two from: - A (device in the) car sends out sound waves (1st) - and the sensor receives the sound waves //after they are reflected back (1) - The device calculates the distance from an object (1st) - by measuring the time between emission and reception of a sound wave/signal (1)	2

3.6 9626/12 Feb/March 2024 · Question 5

Question	Answer	Marks
5	Six from: - The required temperature is set by the user - The required temperature is input using a remote control/touchscreen /number pad - A sensor is used to read/gather/monitor the (current) temperature - The data collected by the sensor is analogue (shows clear awareness that original data is analogue) - The data is converted to digital via an ADC Microprocessor compares this data/temperature to the required/input value (this must be stated and not implied) - If temperature is, nothing happens - If the data is different than the required value, the microprocessor sends a <u>signal</u> ... (1st) ...to the actuator (1) If lower, the microprocessor/actuator switches the heater on If higher, then the microprocessor/actuator switches the heater off - Process is constantly repeated	6

3.7 9626/11 Oct/Nov 2024 · Question 8

Question	Answer	Marks
8(a)	THREE marks available e.g. - Motion sensor detects movement (1) - Infrared sensor to detect heat (1) - Sound sensor to detect level of sound/unexpected sound (may exemplify) (1) - Pressure sensor (placed under a rug/carpet) to detect increase in weight by burglar standing on it (1) - Light sensor to detect a break in a light beam. (1)	3
8(b)	THREE from: - Microprocessor may be programmed to have (a range of) acceptable levels (1) - If sensor is activated/sensor readings are outside the expected levels/range (1) - Data is sent from sensor to Analogue to Digital Converter (ADC) (1) - ADC converts the data to digital (1st) - so can be understood by the microprocessor (1) - It sets off an audible alarm sounds/activates lights/alerts police or alarm company or key holder. (1)	3

3.8 9626/13 May/June 2024 · Question 9

Question	Answer	Marks
9	Six from: • Sensor identifies approach/presence of vehicle (1) • Identifies ONE suitable sensor (Induction loop/pressure sensor/light sensor) (1) • Microprocessor registers presence of vehicle (1) • Microprocessor checks if barrier is raised/lowered (1) • If barrier down it needs to be raised (1) • Microprocessor sends a signal to an actuator/motor (1st) • which then raises the barrier (1) • If barrier up microprocessor does nothing (1) (if mark not awarded for MP 5, a mark may be awarded for “If barrier up”) • Computer checks to see whether vehicle is present/has gone (1) and if vehicle not detected then the barrier comes down (1)	6

3.9 9626/12 May/June 2023 · Question 6

Question	Answer	Marks
6(a)	Infrared • Detects heat (1) • Turns infrared electromagnetic waves into an electrical signal (1) e.g. • Used in a burglar alarm – accept any reasonable use (1) Ultrasonic • Detects sound (1) • Turns sound waves into an electrical signal (1) e.g. • It is used in automated car parking//reversing system (1) • Measure distance (1)	4

Question	Answer	Marks
6(b)	Advantages MAX five from - Increased employment for technicians needed to maintain the systems//maintenance (1) and programmers to program the systems (1) (May also be used as a disadvantage - Award ONCE only) - Data is processed more quickly (1) Reaction times of control technologies is much faster (1) - Computer-controlled traffic lights tend to create fewer traffic jams (1) - Computer-controlled car park barriers lead to fewer/shorter queues (at car park entrances) (1) - Any indication of reduced day to day costs (may be exemplified – e.g. ‘no need to pay anyone to do the job’) (1) - System can operate 24 hours a day//the whole/all day//continuously ... (1) ... without needing to pay overtime (1) - The readings/inputs taken by computers are more reliable/accurate than human readings ... (1) ... the number of car park spaces are more accurately calculated (1) Disadvantages MAX five from - Has increased unemployment for car park attendants (1) (Converse answer may also be given as an advantage – Award ONCE only) - The number of new jobs created tends to be far fewer than the number of old jobs lost (1) (Control systems) can be expensive to install/purchase (1) - A computer-controlled system will not be able to function if there is a problem with the computer/power outage ... (1) ... leading to possible chaos at traffic lights/traffic jams (1) ... causing longer queues at car park entrances (1) - Computer maintenance costs can be expensive (1) Sensors can deteriorate after a period of time //suffer wear and tear (1) ... whereas this would not be a problem with car park attendants (1) Max. six marks if uses a bulleted list.	8

3.10 9626/11 May/June 2025 · Question 4

Question	Answer	Marks
4	One mark per bullet point to a maximum of six marks - They can cause a loss of manual skills (1) - They can stop people from performing simple exercise (1) - They can lead to people becoming lazy (1) - They could become over reliant on microprocessor-controlled devices (1) - The system will not function if there is a problem with the Wi-Fi/internet (1) - The system will not function if there is a power cut (1) - They can be vulnerable to hackers (1) - All devices have to be on the same network (1) - Can increase electricity use (1) - Automatic running of routines may have undesired affect (1)	6

4 Algorithms and flowcharts

4.1 9626/12 Oct/Nov 2024 · Question 4

Question	Answer	Marks
4	Example solution: <pre>graph TD START([START]) --> INPUT1[/INPUT Num/] INPUT1 --> SETMAX[SET MAX to Num] SETMAX --> SETTOTAL[Set Total to Num] SETTOTAL --> SETCOUNT[Set Count to 1] SETCOUNT --> ISCOUNT{Is Count = 21?} ISCOUNT -- YES --> SETAVG[Set Avg to Total/21] SETAVG --> OUTPUT[/Output MAX, Avg/] OUTPUT --> END([END]) ISCOUNT -- NO --> INPUT2[/INPUT Num/] INPUT2 --> SETTOTAL2[Set Total to Total + Num] SETTOTAL2 --> ISMAX{> Num > MAX?} ISMAX -- YES --> SETMAX2[Set MAX to Num] ISMAX -- NO --> INCREMENT[Count = Count + 1] SETMAX2 --> INCREMENT INCREMENT --> ISCOUNT</pre> SEVEN marks available:	7

- 1 Start and End (accept STOP) terminators (1)
- 2 **TWO** Correctly drawn decision boxes with YES/NO decisions (1)
- 3 **All** YES/NO decisions (used) **go to correct outcomes (i.e. all logic is correct)** (1)
- 4 Input box and output box are both parallelograms (1)
- 5 Process box(es) are rectangles (1)
- 6 Two variables (e.g. MAX, COUNT, TOTAL, NUM) initialised/set to zero **before loop** (1)
- 7 Input value from user (attempt at) (1)
- 8 Total set to first input value (1)
- 9 Inputs limited to 20 values (attempt at use of logic to find) (1)
- 10 Sum all values input (1)
- 11 Set new MAX when input value > MAX **in loop**
- 12 Increment COUNT (1)
- 13 **Correctly** calculate average (must include process/calculation) (1)
- 14 Output MAX and average (1) (This may happen anywhere, but if in loop, must be after any new number is added).

4.2 9626/11 May/June 2023 · Question 8

Question	Answer	Marks
8	<pre> graph TD Start([Start]) --> Init[OT ← 0 ET ← 0 count ← 0] Init --> CountInc[count ← count + 1] CountInc --> InputN[/Input N/] InputN --> XCalc[X ← N / 2] XCalc --> IsInt{Is/If X = INT(X)?} IsInt -- No --> OTAdd[OT ← OT + N] IsInt -- Yes --> ETAdd[ET ← ET + N] OTAdd --> Count10{count = 10?} ETAdd --> Count10 Count10 -- No --> InputN Count10 -- Yes --> Print[/Print OT, ET/] Print --> Stop([Stop]) </pre>	8

4.3 9626/13 May/June 2025 · Question 13

Question	Answer	Marks
13	Max five marks from: - Add sum = 0 (sum initialisation) (1) - Add count = count + 1 (iteration of count) (1) - Replace average calculation from * to / (1) - Reverse Yes/No correctly (1) - Add end shape (1) Corrected flowchart: <pre> graph TD Start([Start]) --> Sum0[sum = 0] Sum0 --> Count0[count = 0] Count0 --> Input1[/Input first number/] Input1 --> SumAdd[sum = sum + input number] SumAdd --> CountInc[count = count + 1] CountInc --> Decision{is count = 12?} Decision -- Yes --> AvgCalc[average = sum / 12] AvgCalc --> Display[/Display average/] Display --> End([End]) Decision -- No --> InputNext[/Input next number/] InputNext --> SumAdd </pre>	5

4.4 9626/12 Feb/March 2022 · Question 7

Question	Answer	Marks
7	Eight from: - Input number of numbers - Initialise count - Initialise total - Loop that works involving REPEAT... UNTIL (e.g. count = n) - Input number within loop - Increment count within loop - Update total within loop - Calculation of average - output average	8

	Possible solution could be: INPUT n count ← 0 total ← 0 REPEAT INPUT number count ← count + 1 total ← total + number UNTIL count = n average ← total / n PRINT average	
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4.5 9626/13 May/June 2025 · Question 7

Question	Answer	Marks
7	<i>Example solution:</i> <pre> Count ← 0 REPEAT INPUT mark IF mark > 70 THEN OUTPUT "You have achieved a Distinction" ENDIF IF mark >= 55 AND <= 70 THEN OUTPUT "You have achieved a Merit" ENDIF IF mark >=40 AND <= 54 THEN OUTPUT "You have achieved a Pass" ENDIF IF mark <40 THEN OUTPUT "You have not reached the Pass mark" ENDIF Count ← Count+ 1 UNTIL Count = 20 </pre> Eight marks from: • Initialise Count before Mark variable is input (1) • REPEAT in correct logical location (1) • Input Mark (1) • Correct Distinction grade criterion//correct fail criterion (1) • Correct Merit grade criterion//correct Pass grade criterion (1) • Correct Increment of Count (1) • Increment of Count is within the loop (1) • UNTIL Count = 20 (1) • UNTIL in correct position (1) • Use of one IF...ENDIF (1) • All OUTPUTs correctly match the criteria (NOT awardable if BP4 or BP5 not awarded) (1)	8

Question	Answer	Marks
7	Eight from: 1 mark for each correct statement <pre> count ← 0 INPUT number REPEAT INPUT jobcode, hours CASE (1 - present and location) OF jobcode (1 - allow on FT) 'L' : rate ← 18 'O' : rate ← 22 'M' : rate ← 30 OTHERWISE (1): PRINT/OUTPUT "Invalid job code" ENDCASE CASE and ENDCASE correctly indented wage ← hours * rate statement outside of CASE statement PRINT/OUTPUT wage (present) PRINT/OUTPUT wage (outside of CASE statement) count ← count + 1 UNTIL count = number UNTIL count = number as final statement </pre>	8

Question	Answer	Marks	Guidance																											
2(a)	EIGHT marks available:	8	Each answer is CAO ONLY considers answers on the table																											
	<table><tr><th>Line</th><th>Item to be added</th><th></th></tr><tr><td>80</td><td>4</td><td>1</td></tr><tr><td>90</td><td>Number (accept lower case)</td><td>1</td></tr><tr><td>100</td><td>></td><td>1</td></tr><tr><td>110</td><td>THEN (accept lower case)</td><td>1</td></tr><tr><td>130</td><td>ELSE (accept lower case). Do not award ELSEIF</td><td>1</td></tr><tr><td>140</td><td><</td><td>1</td></tr><tr><td>170</td><td>ENDIF (DO NOT accept lower case or if split)</td><td>1</td></tr><tr><td>180</td><td>Count + 1 (accept lower case)</td><td>1</td></tr></table>			Line	Item to be added		80	4	1	90	Number (accept lower case)	1	100	>	1	110	THEN (accept lower case)	1	130	ELSE (accept lower case). Do not award ELSEIF	1	140	<	1	170	ENDIF (DO NOT accept lower case or if split)	1	180	Count + 1 (accept lower case)	1
	Line			Item to be added																										
	80			4	1																									
	90			Number (accept lower case)	1																									
	100			>	1																									
	110			THEN (accept lower case)	1																									
	130			ELSE (accept lower case). Do not award ELSEIF	1																									
	140			<	1																									
	170			ENDIF (DO NOT accept lower case or if split)	1																									
180	Count + 1 (accept lower case)	1																												
2(b)	FOUR from: - It is the only number that has been entered at that stage (1st) there is no comparison to be made (1) (award once only) - It must be both largest and smallest (1st) there is no comparison to be made//any comparison would be comparing a number with itself (1) (award once only) - Ensures that the highest/lowest mark recorded is valid (1) - Overwriting both set values with a value inputted by the user (1) - One mark for a numerical example of either lower or higher number being ignored (1)	4																												

4.8 9626/12 Feb/March 2026 · Question 7

Question	Answer	Marks
7(a)	SIX marks available, TWO per identified and corrected error: <pre> 10 total ← 10 (1) (Incorrect initial value//value should be 0//value does not need to be set) (1) 20 PRINT "Input number of repeats" 30 INPUT n 40 count → 1 (1) (Count arrow pointing in wrong direction) (1) (Count value should be 0 (this mark may be claimed at line 100)) (1) 50 REPEAT 60 count ← + 1 (1) (Incorrect statement for increment) (1) 70 PRINT "Input number ",count 80 INPUT number 90 total + number ← total (1) (Incorrect order//setting wrong value to variable) (1) 100 UNTIL count = number (1) (Incorrect statement for comparison/should compare to n) (1) 110 average ← total / n 120 PRINT average </pre>	6

Question	Answer	Marks
7(b)	TWO marks available: • A WHILE...ENDWHILE loop will only be executed if the condition is true (1) BUT a REPEAT...UNTIL loop will happen even it is not true. (1) • A REPEAT...UNTIL loop is guaranteed to happen, (1) but a WHILE...ENDWHILE loop may not happen. (1) • For a WHILE...ENDWHILE loop, the condition is tested before the statement runs (1) for a REPEAT...UNTIL loop, the condition is tested after the statement runs. (1) • A WHILE...ENDWHILE loop is a pre-condition loop, (1) and a REPEAT...UNTIL loop is a post-condition loop. (1) • A REPEAT ...UNTIL loop ends when a condition is met (1) but a WHILE...ENDWHILE loop ends when a condition is not met. (1)	2

4.9 9626/12 Oct/Nov 2022 · Question 9

Question	Answer	Marks
9(a)	<pre> PROCEDURE AfterTax(WagesBeforeTax) TaxPaid ← WagesBeforeTax * 35% WagesAfterTax ← WagesBeforeTax - TaxPaid ENDPROCEDURE </pre>	4

	• PROCEDURE and Appropriate name as first statement (ignore any gaps) (1) • WagesBeforeTax in brackets on first line (1) • Calculations and assignments within the procedure result in the correct numerical outcome for the question (Ignore errors with variable names, as long as it is clear which variable is being referred to) (1) • Sensible variable names with no gaps (WagesBeforeTax has been given, must be at least TWO extra variables) (1) • ENDPROCEDURE as last statement (1)	
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Question	Answer	Marks
9(b)	<pre>count ← 0 INPUT NumberOfWorkers REPEAT</pre> • INPUT Hours, Rate (1) (Candidate may do this in two steps) • INPUT Hours, Rate as first line/before BeforeTax is called (allow follow through) (1) • (CALL) BeforeTax(Hours, Rate) (1) • (CALL) AfterTax(WagesBeforeTax) (1) • BeforeTax before WagesAfterTax (1) • PRINT/OUTPUT WagesAfterTax (1) (Must be in correct place) • count ← count + 1 (1) (can be anywhere) • UNTIL count = NumberOfWorkers (1)	6

4.10 9626/11 May/June 2022 · Question 6

Question	Answer	Marks
6	Six from: <pre>REPEAT INPUT temperature IF temperature < preset THEN IF pump switched off THEN send signal to switch pump on ENDIF ELSE IF pump switched on THEN send signal to switch pump off ENDIF ENDIF UNTIL system switched off</pre>	6

1 mark for correct 1st IF statement 1 mark for correctly positioned first action 1 mark for correctly positioned 1st ENDIF 1 mark for correct 2nd IF statement 1 mark for correctly positioned second action 1 mark for correctly positioned 2nd ENDIF 1 mark for correctly positioned 3rd ENDIF 1 mark for correct UNTIL statement... ...1 for correct position of correct UNTIL statement	
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5 eSecurity

5.1 9626/12 May/June 2023 · Question 4

Question	Answer	Marks
4	Eight from: • (because) It is sensitive information about yourself (1) • an example of personal data that could be given (1) plus how this would be detrimental//how used to (e.g.) gain (your) trust (1) Fraudsters could use the data to: e.g. • commit identity theft (1) to (e.g.) buy products (may exemplify) (1) • get past bank security (1st) withdraw large sums of money from bank accounts (1) (for this second mark accept any situation where identity could be proven by details shared online – credit card/inheritance etc) • By sharing (e.g.) phone number may lead to phishing (1) Burglars could use the data to e.g. • target homes of people who announce they are going on away from their home (holiday etc) (1) Employers could e.g. • use comments about politics/religion/their current job//embarrassing photos to decide on future job opportunities (accept any example of an area of opinion, but must be at least one stated) (1) Insurers/employers could e.g. • make decisions based on evidence of risky/dangerous activities (may exemplify about decisions) (1) Max. six marks if bullet list of points	8

5.2 9626/12 Feb/March 2026 · Question 1

Question	Answer	Marks
1(a)	TWO marks available: • Remove geotags//metadata (from the photograph) (1) • Restrict/flag photograph as not for public//change privacy settings (1) • Make access to the social media private (1) • Description of any edit to the photograph to keep personal data secure e.g. ONE from Obscure people's faces/features (1) Obscure features which allow the location to be identified (1)	2
1(b)(i)	FOUR marks available:	4

	- Firewall only allows <u>authorised</u> traffic (DO NOT accept converse – mark as REP) (1) - TWO from: - Use of Packet filtering (1) - Use of Proxy service (1) - Use of Stateful inspection (1) - The firewall controls what can come in AND go out (1) - The firewall operates based on rules (1) - TWO from: IP addresses/IPs of known threats are blocked//only allows access to allowed IP addresses (allow converse) (1) - Only a restricted list of (communication) protocols are allowed (allow converse) (1) - Packages from certain (communication) ports are accepted/denied (allow converse) (1)	
1(b)(ii)	ONE mark available: - Can block legitimate traffic (1) - Can slow down computer performance (accept any description) (1) - IP addresses can be masked/spoofed (1)	1

5.3 9626/12 Oct/Nov 2022 · Question 3

Question	Answer	Marks
3(a)	Two matched pairs from: - Phishing involves sending an email (to gather personal details) ... (1) ... whereas vishing involves a phone/voice call (to gather personal details) (1) (Candidate MUST have referred to personal data/details being gathered by either method within the body of the answer) - Phishing involves sending a website link/email address, asking the receiver to email back/click on link ... (1) ... whereas vishing involves getting the victim to give out their details verbally / leaves a number on their answerphone / tells them over the phone the bank's supposed phone number the victim can call to check the call is authentic (1) - With phishing the link takes the victim to a (fake) website ... (1) ... whereas with vishing the phone number causes the victim to be put through to the hacker's own phone/answerphone (1) - Vishing involves the customer having to respond immediately to the caller ... (1) ... whereas phishing allows the user more time to check whether the email is authentic (1) One mark may be awarded for unmatched statement Two marks may be awarded for one unmatched statement from each side	4
3(b)	Three from:	3

	- Fraudster sends a (cell phone) <u>text message</u> (or VERY close alternative) to persuade customer to divulge their personal information (accept any reference to personal information/data in the body of the answer) ... (1st) ... includes a link to fraudster's/fake/replicated website (1) ... Example of message that could be used (must be more than 'please send information') (1) ... the target's immediate attention/action is demanded (accept any reference to peril or scaring the target into immediate action) (1) - They include a telephone number in the message that connects to an automated voice response system (1)	
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5.4 9626/12 Feb/March 2022 · Question 8

Question	Answer	Marks
8	Four matched pairs from examples such as: It is important to use anti-phishing software often found within web browsers or email software to prevent emails from even arriving... ...but not all web browsers provide this facility so it is very important to use one that does... ...but unless the software is updated at regular intervals it may not be able to prevent attacks Look for grammatical and/or spelling mistakes in emails as this will cause receivers to be reluctant to reply... ...however, it may be possible that users could see minor grammatical and/or spelling mistakes in genuine emails and users may delete them unnecessarily Users should never trust emails that come from people whose names they do not recognise so if any email looks suspicious it is important to delete it... ...but sometimes, emails may come from people whose names they do not recognise but they may still be genuine If an email starts 'Dear customer' rather than using the receiver's name, this can be viewed with suspicion and it is important to delete it... ...however, some companies use this method of communication for general emails so if the emails is not asking for personal information there may be no need to delete it. Emails requesting personal or financial information are likely to need deleting as it is important to know that personal and financial information should never be sent in an email... ... check with sender/company if email asks for personal information as it may be genuine If the email contains a message that the receiver is entitled to a large amount of money it is likely to be fake and deletion is probably needed... ...however, it is possible that a distant relative has left the user a lot of money so it is important to read these messages carefully before considering deletion	8

	Users need to treat links within emails with suspicion... ... but if a user decides never to click on links within emails this can lead to important emails not being acted upon Must be a proper evaluation to obtain full marks Max. four if points not matched Must have expansions to be a proper evaluation	
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5.5 9626/12 May/June 2022 · Question 8

Question	Answer	Marks
8(a)	Four from: <u>Hacker creates</u> a fake website which looks like the actual bank's website/legitimate-looking website Fools the user into thinking they have gone to the correct website/have not been re-directed Hacker installs a piece of malicious software/code on customer's computer/ user downloads a piece of malicious software/code without realising it Software corrupts the Hosts file on the user's computer Software adds the URLs of banks together with IP address of the fake website Fraudster redirects genuine website's traffic to own website/ when user enters a bank's URL they are redirected to hacker's websiteuser is now sending personal details to fraudster's website unknowingly / website asks customer to enter banking information/hacker gains <u>personal</u> information from website /customer types in personal details on website Hacker uses the acquired information from the website to log on to the user's bank account/make purchases/commit identity theft/withdraw cash	4
8(b)	Three from: Use up to date anti-virus software to prevent the downloading of malicious software Use firewalls to prevent malicious software entering the computer Use trusted DNS servers / don't enter personal data into sites without https The latest browsers alert users when a fake website has been loaded/users need to use the latest <u>browser</u> software updates Use a trusted, legitimate Internet Service Provider Check the digital certificate to make sure the site is legitimate Be suspicious and careful not to use <u>websites</u> which contain poor grammar or spelling Avoid using public Wi-fi systems	3

Question	Answer	Marks
6(a)	Four from: • (Self) replicates (1) • A malicious bot infects a host/series of computers (not 'a system') (1st) – takes control (1) – connect back to a central server//bot herder (1) • Automates tasks/provide services that would otherwise be conducted by a human (1) • Bots are used to gather/steal/copy/gather information (1) • Attackers can launch broad-based//flood-type attacks//brute force attacks (against their targets) (1) • They can launch DDoS attacks/spam attacks/relay spam/open back doors on the infected network (1)	4
6(b)	Four from: • Installs a set of tools (1st) – such as e.g. (one of) key stroke logger/web monitoring (1) • Software designed to provide continued privileged/administrator access//remote control (NOT remote <u>access</u>) to a computer (1st) – example of action that can be carried out by the administrator once they have access (e.g.) change system configurations/delete files/change passwords (1) MUST CLEARLY STATE THAT THIS IS ACHIEVED BY THE ADMINISTRATOR/hacker • A rootkit can access log files//spy on the computer owner's usage (for example, key stroke monitoring) (1) • Hides its presence//user is unaware of its presence//installed without computer user's knowledge (1) • Activated/loads before a computer's operating system has (completely) booted up (1) • Root refers to the admin account on Unix and Linux systems (1) • Kit refers to the software components that implement the attack (1)	4
6(c)	Four from: • Spreads from one computer to another (1st) – by e.g. the internet/(infected) external storage device (1) • They can replicate/duplicate (1) • They can delete/corrupt/change data/files/program (1) • They attach/insert themselves to/into another program/file (1) • They can often lie dormant until a particular situation arises (1st) – which causes the computer to execute the code (1) – This can be a specific date/time (1) • Viruses can cause the computer to crash//slow down processing speed/computer system (accept associated concepts) //change a user's homepage/change user's password preventing them from logging on. (1)	4

5.7 9626/12 Feb/March 2023 · Question 9

Question	Answer	Marks
9(a)	FOUR from: • A (standalone) malicious computer program/code (1) • A program that replicates itself in order to spread to other computers (1) • Does not need to attach itself to an existing program/file (1) • Worms consume bandwidth, slowing down the (traffic in a) network (1) • Worms consume processing power, therefore (e.g.) slowing down computer operation/system//stopping the computer from operating (1) • A worm exploits (security) holes in networks/computers (1) • They replicate themselves occupying (more) space (1) • Can modify/delete files//bring other (malicious) software//install backdoor (1)	4
9(b)	FOUR from: • A malicious computer program which is used to hack into a computer (1) • Misleads users of its true intent/disguises itself as authentic software (1) • Enables the person who created it to take control of the computer it has infected (1) • Trojans (generally) do not attempt to infect other files (1) • They do not attempt to self-replicate themselves (1) • Has to be/is activated by (e.g.) the target person (1) ONE mark for any further action a Trojan may perform (1) e.g. • Provide a backdoor onto your computer so a 3rd party can take control • Delete files from the hard disk • Copies data/files • Can corrupt data (damage = TV) • Can be used to download other malware	4

5.8 9626/12 Feb/March 2022 · Question 9

Question	Answer	Marks
9(a)	Three from: Spyware is a type of malware that collects information about a computer user's activities without their knowledge... ...but a user might notice a reduction in processing power and bandwidth Data such as web browsing habits are collected Data such as user's personal data are collected... ...such as the user's credit card numbers/usernames and passwords A keylogger can be used to collect a record of the user's keystrokes Data is sent to the hacker It does not replicate like a virus or worm	3
9(b)	Four from:	4

Hackers can sell the personal/financial information to other criminals Hackers can impersonate the user/commit identity theft Hackers can use the user's financial data to ask the bank for a new PIN Hackers can ask the bank for an extra bank card Hackers can buy goods through internet shopping using the credit card details they have obtained Hackers can withdraw large sums of money from the user's bank account	
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5.9 9626/11 May/June 2026 · Question 7

Question	Answer	Marks	Guidance
7(a)	TWO marks available: e.g. - Will lose access to (all) data/devices/files/system (1) - Financial cost (e.g.) of ransom//checking data when it comes back (1) - Data could be released/leaked/sold (any awareness of data being passed onto a 3rd party) (1) - Business disruption/reduced productivity (1) - Loss of revenue (1) - Liability/compliance costs/risk of prosecution (1) - Reduction in customer confidence (1)	2	

Question	Answer	Marks	Guidance
7(b)	SIX marks available. MAX FOUR if only first mark answers. - Definition of what is meant by physical method of prevention//identifies ONE methods of physical protection (1) Positives MAX FOUR e.g. - Cannot physically access the computers (1) - Physical defences can e.g. list malicious sites//give restrictions on how sites may be used (1st) Therefore, users know (which) sites to avoid (1) - Physical defences can state how external devices can be used (may exemplify) (1st) Therefore, users know if they can be used//how to make them safe (1) - A standalone dedicated computer can be set up for virus checking (1st) So, devices can be checked (1)	6	Do Not Award 'Cannot access' on its own

Negatives MAX FOUR**e.g.**

- Digital attacks can occur even if there is an (e.g.) guard (1)
- Ransomware attacks are more likely to be software based//happen over a network//digital (1st)
Therefore physical methods are ineffective (1)
- Physical defences require the user to follow them//may be a change from working practice (1st)
And so may be ignored (1)
- New staff/temporary staff may not be aware of the restrictions (1)
- Physical restrictions are not automatic//will require staff to make decisions/take actions (1)
- Training may not take place at busy times (1st)
Therefore, it becomes irrelevant (1)

Conclusion (may only be awarded if positive(s) and/or negative(s) are included)

- One mark for a value judgement/conclusion based on two arguments (1)
E.g. it can increase consumer confidence

Question	Answer	Marks
6	One mark per bullet point to a maximum of six marks. • Publish anti-malware policies for staff (1) • Teach employees to report malicious websites//suspicious emails to e.g. the network manager/compliance (1) • Instruct employees to be suspicious about clicking on links (1) • Instruct employees that they must not connect removable media/memory sticks/personally owned device to their computer/the network (1) • Provide a dedicated stand-alone/sandbox computer to check for viruses (1) • Keep operating systems/browser software/anti-virus software/the software up to date (anti-malware sw is the focus of the question) (1) • Install/maintain a firewall (any awareness) (1) • Block employees from downloading software/files onto the computer (1)	6

6 The digital divide

6.1 9626/13 Oct/Nov 2022 · Question 10

Question	Answer	Marks
10	Six marks available: • The gap between people who have access to and the resources to use new information/communication technology and those who don't ... (1) ... This technology can include the telephone, television, personal computers and the Internet (1) • The quality of connection to the Internet may vary between groups/countries (1) • Differing levels of infrastructure between (e.g.) countries (1) • The price of connection to the Internet may vary between groups/countries//ability to afford access to the internet (1) • May refer to inequalities between individuals, households, businesses, or geographic areas (minimum of two mentioned) (1) • The divide between those who have the skills/knowledge/ability to use the technologies and those who do not (1) • The digital divide can exist between those living in rural areas and those living in cities (1) • The digital divide can exist between the educated and uneducated (1) • The digital divide can exist between socioeconomic groups/rich and poor (1) • The digital divide can exist between old and young people (1) • The divide between more and less industrially developed/technologically aware nations (1) • The divide between countries or regions of the world is referred to as the global digital divide (1)	6

6.2 9626/13 Oct/Nov 2024 · Question 8

Question	Answer	Marks
8	EIGHT from, e.g.: Award ANY description of differing access to Information Technology leading to digital divide. ONE mark per description	8

e.g.

- Access to fast broadband can be better for those people living in urban areas than for those living in rural areas (1)
- The quality of connection to the internet may be poorer in underdeveloped countries (1)
- Buying televisions/personal computers/smartphones can be more expensive in rural areas (1)
- Richer people are more/poorer people less able to afford the latest versions of televisions/personal computers/smartphones computers (1)
- Lower paid workers may not be able to afford the hardware to be able to access the IT (1)
- Uneducated people are often not aware of IT/importance of IT/benefits of IT/possibilities of IT (1)
- Young people have grown up with the technology/older people may feel they are too old/unwilling to learn about new technology (1)
- Disabilities of some users may inhibit the use of smartphones/computers. (1)

Question	Answer	Marks	Guidance
7	EIGHT from: - Because there is a digital divide (between them) (1) Possible themes include: - Different locations (1) with different access (1) - Lower disposable income/affordability/socio-economic difference (1) affecting access (1) - Different levels of fear (of any type)/comfort (1) however, this fear will quickly disappear with use (1) if issue is not addressed, this problem will get worse (1) - Different day to day experience/training (1) - Different jobs make different use of IT (1) - Disability (1) - Lack of awareness of possibilities (1) - Any example of choice or lifestyle (absence of or different need lack of interest, lack of online friends (1) Answers may be awarded for: Identification of issue (1) Explanation/description of issue (NOT just another version of the same issue) (1)	8	

6.4 9626/12 Oct/Nov 2024 · Question 5

Question	Answer	Marks
5	SIX from: - Highly educated people are often employed in jobs that offer exposure to IT (1) - Highly educated people will often undertake IT training as part of their education//have IT skills as a result of their education (1) - Highly educated people are often aware of IT/importance of IT/benefits of IT/possibilities of IT (1) - Highly educated people have more access to the higher paid jobs (1st) therefore can afford/access the (e.g.) higher speed internet/most up to date IT equipment (1) (accept converse argument) - People with lower levels of education often work longer hours, therefore have less time to access IT (1) - Lower educated people are less like to invest in IT (1)	6

6.5 9626/12 May/June 2023 · Question 7

Question	Answer	Marks
7	Five from: - Different socioeconomic levels imply differing levels of (disposable) income (1) - Different incomes/socioeconomic groups impact on what people can buy/access/afford (1) such as 'things' ... (must be an extension linked to technology) (1) such as acquisition of skills/paying for training (1) - This impacts on what people can commit to renting (1) - New equipment is costly (1) - Old equipment is not as advanced (1) - People from lower socioeconomic groups tend to have lower digital skills (1) - No/reduced incentive for companies to develop cheaper systems ... (1) ... because the profits would be lower (1) - No/reduced incentive to provide infrastructure to poor areas (1) ... because reduced demand in the area (1)	5

6.6 9626/12 Oct/Nov 2023 · Question 6

Question	Answer	Marks
6(a)	Four from: A general statement of impact: - Inability to find information due to inability to access search engines (or other example of cause. Example must be more than "cannot access the web") (1) - Tasks take longer to complete, which wastes time (1) Up to four marks for four effects and explanations.	4

	Areas include: Reduced health care, e.g. • Inability to get onto a waiting list as cannot access doctor's website (1) • Not aware of imminent health risks as do not have access to a national news service (1) Reduced ability to work remotely, e.g. • Unable to have face-to-face communication over the web, and so cannot work remotely (1) Impact on education, e.g. • Cannot access online learning resources, and so quality of education is lower than could be (1) • Unable to do wider research and so will lack wider understanding of the topic (1) Impact on purchasing, e.g. • Inability to compare prices of products as cannot access search engines (1) • Inability to check on availability of products as can not access search engines (1) Other areas include, but are not limited to: • Impact of limited access to online facilities, banking • Impact of limited access to communication • Impact of limited access to mental health • Impact of limited access to online entertainment	
6(b)	Three from: • Build/improve access to the internet//faster broadband (in rural areas) (1) • (Governments/industry can) provide/improve the needed/broadband infrastructure (for fast broadband in rural areas) (1) • Government provides incentives/subsidise to industry to build infrastructure (1) • Government investigates alternative means of distribution (1) • Create cyber/internet cafes (in rural areas) (1) • Deliver access using satellite technology (1)	3

6.7 9626/12 Feb/March 2024 · Question 12

12	Six available. One for a definition of the digital divide: • The digital divide is the gap between those who have access to//use technology and those who do not Max six from:	6
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	• In this case, the gap is caused by differing levels of knowledge about what technology is available (1) AND differing levels of knowledge about what technology can achieve//benefits to be gained (1) • Not everyone is able to keep up with/keep aware of the constant changes in technology • A person who is unaware of what is available//what can be achieved will not use //not know how to use technology • A person who is aware of what is available//what can be achieved will use technology • Increased digital awareness leads to increased/better infrastructure// more investment in IT (accept personal equivalences) (1) which then leads to increased digital awareness (1) (accept reverse argument for digital unawareness) • This divide may be reduced by education/research/experience • This divide may exist within any socioeconomic bracket • This divide may exist within any age group.	
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6.8 9626/12 Oct/Nov 2025 · Question 13

Question	Answer	Marks
13	Six from: e.g. • Can give differing levels of access to health services (1) • Increased risk of death/length of illness extended (1) • May not have online health services (1) – Reasonable implication (1) • Delayed access to results (1) • May need to attend doctor in person/travel to hospital (1) • Not everyone may be able to book online (1st) – need to visit/telephone (1) – which means time is wasted/used less efficiently (1) • Not everyone can order/reorder medicine/prescriptions online (1) • Doctors can work remotely from the patient (1st) – increased access to doctor/patient (1) – ambulances are less frequently used (1) • Not everyone can submit/receive test results online (1) • Not everyone can be monitored remotely (1)	6

6.9 9626/12 May/June 2024 · Question 3

Question	Answer	Marks
3	Six from, e.g.: One mark for a definition of the digital divide: e.g. • The digital divide is the gap between those who have access to//use modern/ICT technology and those who do not (1)	6

	• Improve communications coverage/access to the internet/access to broadband (1) • Improve infrastructure (1) such as (e.g.) invest in satellite technology/build more mobile phone masts/lay fibre cabling (1) • Provision of public access to internet facilities in, e.g. (ONE OF) government buildings/libraries/schools and internet kiosks/cafes (1) • Provide/subsidise laptops/PCs/technological equipment to, e.g. (ONE OF) the elderly/the poor/school children (must be a specific group likely to be negatively affected by digital divide) (1) • Provide/subsidise of mobile or cell phones to, e.g. (ONE OF) the elderly/the poor/school children must be a specific group likely to be negatively affected by digital divide) (1) • Provision of training services/sessions (1) • Subsidise access to a named service (1)	
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6.10 9626/11 and 9626/13 Oct/Nov 2025 · Question 14

Question	Answer	Marks
14	Four marks for two matched group-remedy pairs: e.g. Groups (1st) • Old people (1) • People in rural areas (1) • Low/different levels of education (1) • Lower socioeconomic group/people who cannot afford to purchase (1) • People with learning difficulties (1) • People with physical impairments (1) • People with sensory impairments (1) • People in less industrially developed nations (1) Remedies (1) • Education/training (1) • Drop-in centres support (1) • One to one support (1) • Invest in infrastructure (1) • Provide specialist interfaces (1) • Provide (cheap) loans/subsidies/price ceiling/free (1) • Recycling/refurbish/repurpose equipment (1)	4

7 Expert systems

7.1 9626/11 Oct/Nov 2024 · Question 10

Question	Answer	Marks
10	EIGHT from: Max five if only first points • Knowledge base comprises a database of facts and a rules base (1st) – It is the information used to make the decision (1) – Database of facts gathered from experts (1) • Rules base is a set of rules/statements (1st) – usually of the form IF . . . THEN//description of IF ... THEN in practice in context (outcomes based on what is seen/observed/measured) (1) • The shell includes the user interface, explanation system, inference engine and knowledge base editor (THREE FROM) (1) • Inference engine is the reasoning part of the system (1st) – That uses forward chaining or backward chaining (1) • User interface is how the computer interacts with the user, (1st) – displays questions/information (1) – enables the user to type in answers to the questions (1) • Knowledge base editing software/knowledge base editor enables the knowledge engineer to edit rules and facts (1) • Explanation system explains to a user the chain of reasoning used to arrive at a particular conclusion. (1)	8

7.2 9626/12 Oct/Nov 2025 · Question 3

Question	Answer	Marks
3(a)	Four from: • Acts as a reasoning system/applies logic/makes decisions (1) • Searches the knowledge base/database (1) • Selects/applies/uses applicable/relevant rules (1) • Uses rules to interrogate the knowledge base/database (1) • Uses the rules to determine outcomes (1) • Uses IF THEN analysis (1) • Uses forward chaining/backward chaining to reason/arrive at solutions (1) • Request clarification from the user if necessary/asks (further) questions from the user if necessary (1)	4
3(b)	Four from:	4

	• Data is collected from experts/relevant sources/data mining (any awareness of source) (1) • Relevance of data is decided (1) • Data is entered (into knowledge base) (1) • Uses a knowledge base editor to edit/created by a knowledge base engineer (1) • Design/decide the rules base (1) • Create the rules base (1) • Create the links between the rules and knowledge/data (1)	
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7.3 9626/13 May/June 2024 · Question 3

Question	Answer	Marks
3	Six from: Forward chaining MAX FIVE • Forward chaining starts with/uses the data (1) • Uses inference rules to extract more data. (1st) • until a goal is reached (1) • An inference engine searches until it finds an IF statement known to be true (1) • The inference engine uses the 'THEN' part of the IF THEN statement to cause the addition of new information (1) • Inference engines will iterate through this process (1st) • until a goal is reached (1) • This method is called data-driven (1) Backward chaining MAX FIVE • Backward chaining starts with/uses a list of goals/hypotheses (1) • Uses inference engine to search for data that fits a goal/hypothesis (1) • An inference engine searches until it finds a THEN part that matches a desired goal (1) • If the IF part of that rule is not known to be true (1st) • then it is added to the list of goals (1) • This method is called goal-driven (1)	6

7.4 9626/13 May/June 2023 · Question 10

Question	Answer	Marks
10(a)	Forward chaining	1
10(b)	Four from: • An inference engine searched the rules until it found one where the IF statement was true (1) • An inference engine rejects any rule that is not TRUE (1) • The IF Hortense is an Ostrich is NOT TRUE (1) • The IF Hortense sings and Hortense is 15 cm is TRUE (1) • The 'THEN' part caused the addition of new information (1) • Inference engine looked for another matching IF... (1) • The IF X is a canary - THEN X is yellow is TRUE (1) • Inference engine stopped because the goal was reached (1)	4

7.5 9626/12 Feb/March 2026 · Question 6

Question	Answer	Marks
6	FIVE marks available: • Use of an example (1) FOUR from: • It is part of/uses forward chaining (1) • A process of using/analysing data to find a solution (this mark may be awarded for a description of this process as part of the description of the context) (1) • Data entered into the system determines the rules to be used (1) • (The inference engine) starts with the data and progresses towards a goal (1) • (The inference engine) will try to match the (inference) rules//searches the rules (award any indication that the rules base is interrogated) (1st) Until one matches the IF statement (1) • The THEN part of the rule brings out new information/data/advice (may answer in context instead of saying information/data/advice) (1) • Process is repeated until a goal is reached//no further possible solutions can be found (1)	5

7.6 9626/13 Oct/Nov 2024 · Question 7

Question	Answer	Marks
7	FOUR from: • It is used in backward chaining (1) • It starts with a goal/set of goals (1st) – which establish the rules to be followed (1) – to identify initial conditions (1) • Inference engine uses the rules base (1) – to find one which has a THEN part that matches a required goal (1) (Award once only) • Inference engine explores the rules (1st) – until it finds one which has a THEN part that matches a required goal (1) (Award once only) • If the IF part of that rule is known to be true (1st) – it is added to the list of goals. (1)	4

Question	Answer	Marks
9	Eight from: • Identification of three of: knowledge base, explanation system, knowledge base editor, inference engine, user interface, rules base (1) • Inference engine is able to find possible diagnoses by using a form of reasoning (1) • The inference engine uses the data or facts in the knowledge base to reason through the symptoms (1) • (User interface/expert) system asks questions/for symptoms about the illness (1) • Answers//symptoms/required data are typed in (1) • Inference engine compares symptoms to those in the knowledge base (1) • Any awareness of use of IF...THEN... rules/comparisons//rules base consists of IF...THEN rules... (If and THEN both used in the answer) (1) • Searches the inference rules to find one where the IF statement is known to be true (1) • If answer not found, further questions are asked (1) • When such a rule is found, the inference engine uses the 'THEN' part ... (1) ... to find the illness (1) • Possible/suggested/a range of diagnoses are output (to user interface) (1) • Explanation system produces reasons for suggestions/diagnoses (1) • Reasons for suggestions output using the user interface (1) • Doctor can choose to use or reject answer (1)	8

Question	Answer	Marks	Guidance
2	EIGHT marks available: • A user interface is used to input (1) • IF....THEN (any awareness of USE of) (1) • Any awareness of the use of rules (may exemplify) (1st) These rules are stored in the knowledge base (1) • An expert system takes prior learning/knowledge and applies it to a problem (1) • Locations/address (for delivery) are input (1) • Details of delivery are input//examples of details (weight/dimensions etc) (1) • The responses will be compared with the facts (1st) By the inference engine (1) In the knowledge base (1) • Suggested routes (1) and estimated times of delivery will be output (1) To the user interface//to a printed list (1)	8	Do Not Award • One word answers – must be the USE of. • ANY answers that are not in the scenario • GPS (references to real time processing, traffic delays etc may be taken as GPS) • Parcel collection The question states ‘in this scenario’ therefore, answers have to be in this scenario

Question	Answer	Marks
5(a)	Four from: • A user interface so questions/queries about (e.g.) financial details appear (1) • A user interface so answers to questions can be input (1) • Knowledge base editor to enter/amend (e.g.) current financial facts and figures (1) • Database of facts/knowledge base to hold/store current financial details (1) • Rules base to hold/store financial rules, (1) • Inference engine to interrogate knowledge base//uses forward/backward chaining to come to a conclusion (1) • Explanation system to describe why/say why/justify how the conclusions were arrived at. (Do not award “explains”) (1)	4
5(b)	Six from: • Reduced staff costs (1) • Saves staff time in writing (e.g.) documentation (1) • Makes bank more competitive (1) • Expert systems give better advice than humans (1) • Increase customer satisfaction (1) • Makes the bank look more reliable (1) • Expert system, does not need e.g. breaks/holidays (1) • Cheaper to copy an expert system than train new staff (1) • Errors are reduced. (1)	6
5(c)	Three from: • It can lead to bad reviews/reputation (1) • It can lead to loss of customers (1) • Expert systems cannot apply common sense (1) • Expert systems cannot be creative in//cope with unusual situations (1) • Costs of creation//initial installation//initial purchase is high (1) • Knowledge base/rules require frequent updating//knowledge goes out of date (1) • Need to pay/employ a (e.g.) technician//pay maintenance costs for the expert system (1)	3

7.10 9626/12 May/June 2026 · Question 7

Question	Answer	Marks	Guidance
7(a)	TWO from: - Knowledge base editor (1st) allows the engineer to update the knowledge base (1) allows the engineer to edit the rules base (1) - Knowledge base (1st) stores all the facts/information (1) stores the rules base (1) - Inference engine (1st) Applies logic (1) Uses IF THEN reasoning (1) (Uses reasoning to) identify possible solutions/conclusion (1) Uses forward chaining/backward chaining (1) - Rules base (1st) Contains the IFTHEN reasoning (that is relevant to the expert system) (1) - Contains the FORWARD/BACKWARD chaining to reach a solution/conclusion (1) Contains the rules that the expert system uses to draw conclusions (1)	4	Do Not Accept: - User interface systems - Explanation systems
Question	Answer	Marks	Guidance
7(b)	EIGHT marks available: The final decision to follow the advice given by the expert system rests with the mechanic (this may be presented as a positive or a negative aspect) (1)	8	

	Benefits Max 6: e.g. - Provides a consistent analysis (1) - Knowledge is based on input from a range of experts (1st) therefore is based on people who have solved similar issues (1) ONE FROM as (1st) Uses a process of reasoning based on facts and rules Uses what-if analysis Uses IF THEN analysis so there is a structured process of reasoning (1) - Produces a range of possible solutions (awareness of more than one) (1st) based on answers to a set of questions (1) - Provides full coverage//covers all areas/issues (1) - Includes areas about which the mechanic does not know a great deal (1) - Provides an explanation for decisions (1st) therefore the mechanic can learn (1) - Available at all times (1st) therefore do not need to contact another mechanic at an inconvenient time (1) Drawbacks Max 6: e.g. - Can only deal with problems that are recorded in the expert system (1st) therefore anything outside of the scope/experience of the experts/knowledge base cannot be dealt with (1) - May need to create a new expert system//update current expert system (1) As new cars/engines come onto the market//new technology is used in cars (1)		
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Question	Answer	Marks	Guidance
7(b)	<u>If questions are badly structured</u>, will not have all the information needed to solve the problem (1st) therefore suggested solutions may be irrelevant/damage the car further (1) Conclusion (may only be awarded if positives(s) and/or negatives are included) - One mark for a value judgement/conclusion after the discussion of at least two arguments (1)		

8 Spreadsheets

8.1 9626/12 May/June 2026 · Question 3

Question	Answer	Marks	Guidance
3(a)	TWO marks available: • (Data) import from a different source/spreadsheet/database (1) • User forms which provide input boxes/radio buttons (1) • Macros to run input boxes (1) • Drop down lists from which can select an answer (1) • Input boxes linked to specific cells (1) • Copy and paste from another source/spreadsheet/database (1) • Autofill populate cells based on a set pattern/repeated data (1) • Fill series populate cells based on a set pattern (1) • Radio buttons to select options (1)	2	

Question	Answer	Marks	Guidance
3(b)	FOUR from: • Spreadsheet software allows what-if analysis to be carried out (award any mention of what if analysis) (1) • Can sort data into a rank order (concept of ordering) (1) • Can use macros which can create routines /subroutines (1) • Macros/subroutines can carry out fixed routines//e.g. goal seek (1) • Conditional formatting can show/highlight data ONE FROM that matches//is greater than//less than criteria (1) based on rules/conditions (may exemplify) (1) • Carries out calculations such as subtraction (accept any example of a calculation) (1) • Formulae can be created for calculations (1) • Functions can perform set tasks OR (one mark in total) Uses Functions such as • OR (one mark in total) Identifies a function by name and states what it does (1)	4	Award first four attempts only Do Not Award: Features which are NOT directly to do with analysis

- | | | |
|---|--|--|
| <ul style="list-style-type: none">• Charts/graphs to present values//allow for comparison//visualise data (1)• Pivot Tables to analyse large quantities of data in a visual form (1)• Filtering based on set criteria (may exemplify e.g. for a certain class) (1) | | |
|---|--|--|

8.2 9626/13 May/June 2024 · Question 4

Question	Answer	Marks
4	Four from: Max two for each: <i>Absolute:</i> - Always copies the same cell reference without modification when replicated/a fixed reference to a cell (1) - Any example of an absolute value/formula/source cell not changing (1) <i>Relative:</i> - It copies the cell reference, but will increment the cell reference when replicated//cell reference changes as the formula location changes (1) - Any example of a relative value/formula/source cell changing (1)	4

8.3 9626/12 May/June 2025 · Question 2

Question	Answer	Marks
2(a)	Max 4 from: VLOOKUP - It takes the B from cell D12 (1) - It looks for B in cells D5:E8 (1) - Returns/finds the corresponding value//20%//discount percentage (1st) - from column E/2nd column from the range (1) - It looks for an exact match//If it does not find an exact match it will return an error (1) Max 2 from: CALCULATION - It multiplies the percentage discount by the Price (1) - Subtracts discount from the original price. (1)	5
2(b)	Four marks from: Both are needed because some cell references need to remain the same (1) whilst others change (1). Absolute cell referencing always refers to the same cell therefore does not change (1st) - Valid example of a cell reference that must not change (1) Relative cell referencing changes depending on the position in the spreadsheet//may exemplify new positions (1st) - Valid example of a cell reference that must change. (1)	4

Question	Answer	Marks
6	THREE from: • It will check the contents of cell B6 (any awareness that outcome is based on contents of B6) (1) • If the contents are NOT less than 5 (1) OR NOT greater than 8 (1) (accept converse) – ONLY award following if both above stated (NB note effect if answer above is converse) – The cell displays 4th Quarter (1) – if not/otherwise the cell displays a null string. (1) (Answer is dependant) - OR • If condition not met, a blank cell is displayed (1)	3

Question	Answer	Marks	Guidance
3(a)	TWO marks available: - Use a drop-down menu which only allows selection from the list (1) - Accept only whole number e.g. between two extremes/greater than X (1) - Accept only decimal e.g. between two extremes/greater than X (1) - Accept only date e.g. between two extremes/equal to a set date (1) - Accept only time e.g. between two extremes/before a set time (1) - Restrict text length e.g. between two extremes/greater than X (1) - Set allowable value to be equal to that in another cell (1) - Use a (run on change) macro to check the value (1)	2	Do Not Award 'Use validation tools' - Length check on its own, but allow description of length check
3(b)	ONE mark available: A cell reference that will not change when the formula is dragged down//copied to another cell/replicated (1)	1	MUST have both parts
Question	Answer	Marks	Guidance
3(c)	TWO marks available for one complete explanation Identify benefit (1st) state why it is a benefit/state the beneficial implication (1) - Identification of benefit (1st): e.g. Fewer errors Reduced time to create the spreadsheet/formula - Explanation (1) e.g. Because formula only has to be typed once Because can enter formula in one cell and can (e.g.) drag down Because cell references automatically change as copied/dragged/replicated	2	Do Not Award - Makes data more accurate, however, if this is further explained – such as fewer errors, you may award fewer errors - Enter numbers and drag down – has to be formulas
3(d)	SIX marks available:	6	

	• Checks value from cell C2 (1) • If the contents are Dog (1) • The first IF statement is run (1) • It will check the contents of cell D2 (1) • If the contents are Male (1st) It will display "Block A" (1) If not, it will display "Block B" (1) • If C2 not Dog (1) • It will check contents of cell D2 AND if the contents are Male (1st) It will display Block C otherwise it will display Block D (1)		
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8.6 9626/11 May/June 2025 · Question 12

Question	Answer	Marks												
12	One mark per bullet point to a maximum of six marks. <i>Normal</i> - Data is accepted by the system - Any integer between 0 and 60(inclusive) <i>Abnormal</i> - Data outside of the range which will be rejected/not accepted by the system - Any data that is not an integer between 0 and 60 <i>Extreme</i> - Data that is at the boundaries of the acceptable range 0 or 60 E.g. <table border="1"> <thead> <tr> <th>Type of test data</th><th>Example of mark entered</th><th>How the validation rule deals with the mark entered</th></tr> </thead> <tbody> <tr> <td>Normal</td><td>41</td><td>Data is within the range data and will be accepted by the system</td></tr> <tr> <td>Abnormal</td><td>68/-2/pink</td><td>Data outside of the range which will be rejected/not accepted by the system</td></tr> <tr> <td>Extreme</td><td>0/60</td><td>Data that is at the boundaries of the acceptable range and will be accepted by the system</td></tr> </tbody> </table>	Type of test data	Example of mark entered	How the validation rule deals with the mark entered	Normal	41	Data is within the range data and will be accepted by the system	Abnormal	68/-2/pink	Data outside of the range which will be rejected/not accepted by the system	Extreme	0/60	Data that is at the boundaries of the acceptable range and will be accepted by the system	6
Type of test data	Example of mark entered	How the validation rule deals with the mark entered												
Normal	41	Data is within the range data and will be accepted by the system												
Abnormal	68/-2/pink	Data outside of the range which will be rejected/not accepted by the system												
Extreme	0/60	Data that is at the boundaries of the acceptable range and will be accepted by the system												

8.7 9626/13 May/June 2025 · Question 5

Question	Answer	Marks
5	One mark per bullet point to a maximum of six marks. Valid/Normal//Data that is from within the acceptable range (1st) - To test that the output is as expected/valid (1) - To test that it is accepted (1) Invalid/Erroneous//Data//Abnormal that is outside of the acceptable range (1st) - Data is of the wrong type/invalid (1) - To test that it is rejected (1) Boundary/Extreme//Data that is at the boundary (of the acceptable range) (1st) - To test that the boundaries are set correctly (1) - To test that the outcome is as expected – accepted and correct output (1)	6

Question	Answer	Marks	Guidance
8(a)	TWO from: - In bold (1) Centrally aligned//centred horizontally/vertically (1)	2	Do Not Accept Border – not a content
8(b)	THREE marks available: Normal test data (max ONE) - Data that should produce expected results (1) - Data that is within acceptable boundaries//correct range (1) - Should be accepted by the cells (1) - Sensible data (1) Extreme test data (max ONE) - Data that is at the boundary (1) - At the limit of acceptability (1) Abnormal test data (max ONE) - Falls outside acceptable limit/range (1) - Data that is invalid (1)	3	
8(c)	THREE marks available, one per type of test: Normal - Any single number between 44 and 50 (inclusive) (1) Extreme 44//50 (ONE ONLY – NOT BOTH) (1) Abnormal - Any other single number beyond range 44 - 50 (1) - Any data other than numeric (1)	3	Accept real values between 44 and 50 Answer must be actual data (e.g. 50) and not description of data (e.g. 'a value between 44 and 50'). If descriptions given, do not accept this as non-numeric for abnormal test data

Question	Answer	Marks	Guidance
8(d)	FIVE from: =IF(F3>C3,"Check Score",(IF(F3>B3,A3,(IF(F3>B4,A4,(IF(F3>B5,A5,A6))))))) - The formula will not produce the correct grade for all scores (1) at the (lower) boundary of each grade (1) it will produce the grade below (1) Wrong inequality symbol marks (route 1) - The formula only uses an inequality symbol (1) and should have used an inequality (accept equivalent statement, such as > has been used) AND an equals statement (1) - Therefore only marks above the boundary are awarded the grade (1) - An (e.g.) A grade piece of work, gets a B grade. (1) Wrong number marks (route 2) - The wrong value has been used as the lower boundary (1) - The formula uses a greater than inequality (1) the value from the top of the lower boundary should be used (1) - Therefore only marks above the boundary are awarded the grade (1) - An (e.g.) A grade piece of work, gets a B grade. (1) One mark may be awarded for: - One mark for showing how the formula may be improved:	5	Marks may be awarded from route 1 OR route 2 only. Where candidate mixes both explanations, mark to the candidate's best advantage.

Question	Answer	Marks	Guidance
8(d)	e.g. • IF(F3 > B3 – 1 • IF(F2 = B3 etc. IF(F2 > 43		

8.9 9626/12 Feb/March 2025 · Question 7

Question	Answer	Marks
7	FOUR marks available. TWO per matched answer. - Identify a suitable feature of a spreadsheet (1) - State how this feature can be used for the purpose (1) E.g.: Award first answer in each section only, unless no answer written in Way 2. - Manually store/save data about a student's scores/progression (1) to create a record of the student's performance. (1) - Automation/Formulas could be implemented in a spreadsheet (1) to calculate/predict averages of a given/specific time frame. (1) - Conditional formatting could be used in a spreadsheet (1) to highlight grades that meet a set criteria/condition. (1) - A graph/chart could be created from data (1st) - To represent/visualise a student's progression overtime in any given time/test. (1) - A complex/advanced spreadsheet with formulas that function across sheets could be created (1st) - To compare results/scores/progression/trends across different students/groups/classes/tests. (1) - Sort scores into date order (1) so progress over time can be seen (1)	4

8.10 9626/11 May/June 2025 · Question 2

Question	Answer	Marks
2	Max 4 marks: MUST be a comparison <i>Benefits:</i> Easier to create a storage systems//multiple tables/worksheets (1) Easier to calculate (1) Easier to create complex formulae//carry out complex calculations (e.g. use functions (if state functions, this is equivalent to complex)) (1) - Repeated data can be easier to enter (1) - Charts are easier to produce (1) - Includes conditional formatting to (e.g.) highlight data easier (1) Max 4 marks: <i>Drawbacks:</i>	6

- | | |
|---|--|
| <ul style="list-style-type: none">• Information may exist across many sheets (1)• Large spreadsheets are more difficult to use/read (1)• Spreadsheets are best suited to single focus (e.g. per year) (1)• Formulae may need to be modified as spreadsheet grows (1)• Harder to create complex queries (1)• Harder to create a report (1)• Validation is easier in a database (1)• Repeated data exists in spreadsheet//databases are normalised (1)• Referential integrity reduces (e.g.) redundant data (1) | |
|---|--|

9 Modelling

9.1 9626/12 Oct/Nov 2024 · Question 6

Question	Answer	Marks
6	SIX marks available e.g. • Computer is used to create the rules behind a simulation (1st) so that the simulation matches the real world (1) • Easier to try out a range of different scenarios//allows what-if analysis (1) • It is less costly than if a mistake is made in real life/it is costly to remedy mistakes (1) • More difficult to construct several designs in real life (1) • Can be used to check on/make predictions for complicated scenarios//scenarios with a lot of variables (1) • Can observe aspects that would be difficult to observe in real life (1) • Can save time (1st) more tests can be done in a shorter amount of time (1) a much greater number of tests can be performed (1) • Simulation modelling can be used to plan for infrequent/one off events (1) • Computer models are much safer than experimenting in real life (1) • Faster, can speed up variables that would impact on results e.g. climate change (1)	6

9.2 9626/12 Feb/March 2026 · Question 8

Question	Answer	Marks
8	SIX marks available: ONE mark per identified characteristic, ONE mark for the description/purpose of that characteristic. - Can carry out WHAT-IF analysis (1) So that scenarios can be explored (may exemplify scenario being changed)(1) Allows for complexity (1) - Has inputs/outputs (1st) So that values can be set by the user//so that results can be seen by the user (1) - Can create rules (1st) These define how the inputted data/values are used//the calculations that take place (1) - Can edit rules (1st) To take account of changing parameters/conditions (1) - Cell protection (1st) Protects cells from being (accidentally) changed by the user (1) - User forms (1st) Allows the easy input of data (1) Allows data to be entered in an organised way (1) - Can create macros (1st) To automate processes (1) - Automatic recalculation (1st) Outputs automatically updated (1) - Can hold/handle large amounts of data (use of Multiple – award as BOD).(1st) To allow for different possible outcomes/scenarios (1) Allows for complexity (1) (may be repeated from MP 1) Note: – NOT DEPENDANT - Includes function/formulae (1) to perform calculations/carry out equations (1)	6

9.3 9626/12 Oct/Nov 2022 · Question 4

Question	Answer	Marks
4(a)	Three from: - They could increase/change (the number of customers by increasing/changing) the value in <u>cell D1</u> ... (1) - They could increase/change (the number of snacks by increasing/changing) the value in <u>cell C6</u> ... (1) (accept decrease answer) ... - They could increase/change (the number of drinks by increasing) the value in <u>cell C5</u> ... (1) (accept decrease answer) ... - They could increase/change (the cost of souvenir supplies by increasing) the value in <u>cell D17</u> ... (1) (accept decrease answer) The value in D21/profit would change (showing the new profit/loss//profit would change) (1)	3
4(b)	Four marks available	4

Max **three** from:

- The organisers would select the cell **containing the formula** that returns the result they want/D21, **and select goal seek** (1)
They input the value they want the formula to return/5000 (1)
They enter the cell reference of the input value that the spreadsheet can change to reach the target/D1 (1)

Max **one** from:

- The number could be to several decimal places and needs to be a whole number of concert goers (1)
- If other assumptions are incorrect (e.g. the number of snack/drinks bought per concert goer), the number produced by goal seek would be inaccurate (1)
- Venue may not be able to hold that many people (not 'number too large' on its own) (1)

9.4 9626/11 May/June 2026 · Question 4

Question	Answer	Marks	Guidance
4(a)	FOUR marks available, two marks per matched and described example. • What-if analysis explores what will happen if something else occurs/didn't occur/a variable is changed (1) • Rate of birth can be changed (1) to see impact of increased babies (1) • Rate of mortality can be changed (1) to see impact of increased//decreased teenagers//young adults/adults on number of births (1) • Life expectancy can be changed (1) to see impact of people living longer//shorter on overall population. (1) • Change structural constructs (e.g. schools, hospitals, access to water) (1) to see impact on e.g. mortality rates (1)	4	
Question	Answer	Marks	Guidance
4(b)	EIGHT marks available: Simulations can model/represent what happens in the real world/a natural disaster (1)	8	

Positives (max 6)

- Simulation can be used in advance of a disaster (1)
- Do not have to wait for the disaster to happen//simulations can run now (1)
- Simulations can show (possible) impacts/damage of the disaster (1)
- Identifies aspects that can be planned e.g. evacuation routes (1)
- (e.g.) buildings/infrastructure can be checked to check resilience (1)
- Structural/environmental changes can be made (based on the results) (1)
- The replacement cost of a e.g. collapsed bridge exceeds the cost of a simulation (1st)
Therefore money is saved by using on a simulator (1)
- Simulations can **repeat** scenarios//allow for trial and error//can test different plans out (1st)
e.g.
So the response team can explore different responses//check if their response works (1)
Learn from their mistakes (1)
- There is no risk to human life in a simulation (concept of reduced risk/not being in a risk situation) (1st)
Therefore the simulation is safer (1)
- Small changes can be made to the scenario//different scenarios can be checked/explored/used (1st)
To research into marginal effects (concept of) (1)
To allow the team to practice in changing conditions/explore the impact of small changes (1)
- Simulations can be recorded (1st)
for (e.g.) future analysis/training (1)

Question	Answer	Marks	Guidance
4(b)	Negatives MAX 6 • Simulations may not include all possible events (1) Simulation is not realistic (1) • Simulations assume fixed responses/fixed rules (1st) People may respond differently in real life (1) • Initial cost of purchase of software may be expensive (1) • Interpretation of the outcomes requires experts//team members from different groups (1) • Interpretation of the outcomes requires time//has a cost implication (1) Conclusion (may only be awarded if positives(s) and/or negatives are included) • One mark for a value judgement/conclusion based on two arguments (1)		

9.5 9626/13 May/June 2025 · Question 11

Question	Answer	Marks
11	Eight marks from: MAX four for 1st marks only <i>Evaluations: E.g.:</i> • Can interrogate/interact with databases to extract data (1st) – So data only has to be typed/entered in once (1) • Can only use the features of the spreadsheet (1st) – Therefore model may be limited (1) • Spreadsheet software may include unneeded features (1st) – Which could be a distraction//which need to be hidden//increase file size (1) • Spreadsheets include formulae//functions (1st) – So complicated calculations can be created more easily (1) – Reduced chance of error (1) • Calculations can be performed more quickly/more easily/recalculated automatically (1st) – so spreadsheet can be used by someone who does not spreadsheet editing skills (1) • What-if statements can be asked/edited (1st) – without rebuilding the model/spreadsheet from scratch each time (1) • Creates graphs (1st) – Which allows data to be more easily interpreted (1) • Graphs that are produced will automatically change (1st) – as new values are added/old values altered (1) • Models/spreadsheets provide objective results (1st) – not affected by user's bias (1) • The use of templates provides consistency over time (1st) – therefore do not need to relearn the model (1) • Valid conclusion drawing together/based on arguments presented in the answer (1)	8

9.6 9626/12 Oct/Nov 2025 · Question 4

Question	Answer	Marks
4	Four available. Max two for each matched pair: e.g. • Impossible/difficult/improbable to include all the possible variables (1st) – too many/unknown/unexpected variables (variables in widest sense) (1) • Spreadsheet software is limited size/capacity/scalability (1st) – plus suitable explanation of an issue this causes (1) • May not have the required formulae/functions (NOT “options”) (1st) – plus suitable explanation of an issue this causes (1)	4

	• Need training to use the software (properly)/requires an expert/complicated for new users (1st) – plus suitable explanation of an issue this causes e.g. cost of expertise (1) – therefore takes time to create (do not award as extension for “requires an expert”) (1) • Spreadsheets can be edited by anyone who has the software (1st) – So different versions may develop (1) – So different outcomes arrived at (1)	
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9.7 9626/13 Oct/Nov 2023 · Question 11

Question	Answer	Marks
11	Eight from: • Weather forecasting used to involve several weather forecasters and days of calculations (1st) • today one model can manage the task (1) • Computer models help forecasters to make more reliable predictions (1) • Models can be used to predict future patterns (of weather) (1) • More complex models create more accurate forecasts (1st) • which allow forecasts further into the future (1) • Human weather forecasters are still needed to evaluate the results (1) • Weather forecasting requires the use of a lot of data (1) • To be effective models must consider many variables (1st) • such as (TWO Variables identified): atmospheric pressure, humidity, rainfall, temperature, wind speed, wind direction (1) • as well as calculations such as (ONE of): vertical air motions, temperature changes, moisture processes (1) • A lot of computing power is needed to run a computer model so supercomputers have to be/are used (1) • Supercomputers are (very) expensive (so makes the process an expensive one) (1) • Models can be used to compare current conditions with previous similar conditions/shown trends over a period of time (1) Max 6 marks for just bullet points	8

Question	Answer	Marks
9(a)	Four from: • Definition of a simulation e.g. Simulations are a digital representation of a situation as it would happen in real life (1) • Using computer models avoids safety problems (1st) • such as e.g. explosions/meltdowns (1) • Using computer models cuts costs (1st) • as don't have to pay as much money for workers//materials to replace damaged reactors (1) • Can obtain results in a short period of time regarding reactions that take a long time in real life (1) • Re-designing computer models is cheaper than re-building a nuclear power plant (1) • Can simulate scenarios that are hard to observe in operating power plants//in real life (1)	4
9(b)	Four from: • Simulation may not be 100% accurate (1) • Models can't always recreate exactly the real-world experience/may miss variables (1) • Very complex to develop a simulation (1) • Specialist knowledge is required to build model (1st) • which requires specialists which costs money (1) • Specialist knowledge is required to analyse results (1st) • which requires specialists which costs money (1) • There may be too many variables to include in the model (1st) • to exactly simulate the environment/design/working of the nuclear reactor (1) • leading to inaccurate results (1)	4

9.9 9626/11 May/June 2024 · Question 6

Question	Answer	Marks
6	Eight from: Advantages: MAX SIX - The running/maintenance costs of a flight simulator are less than an actual aircraft (1) - Avoids repair costs for real aircraft (1) Different/various/a range of situations can be simulated in a flight simulator (1st) - planned at the start of the simulation (1) - during the simulation (1) - Events can be simulated without waiting for them to happen in real life (1st) - such as (suitable named example) (1) - Situations can be repeated multiple times until the pilot gets it right (1) (Safer because) Situations can be practiced in a flight simulator without putting the humans/trainee/flight training instructor in danger (1) - Flight simulator does not need to be at an airfield//can be in a convenient location (1) - Can concentrate training on specific controls (1) - The environment benefits from flight training in a simulator (1st) - as there is no pollution created by a flight simulator (1) - The level of noise in a flight simulator is not as loud as in an aircraft (1st) - makes the communication between the instructor and trainee much easier (1) Disadvantages: MAX FIVE - Simulators are not as realistic/real as a real aircraft (1) - Pilots get overconfident and may compromise safety (1) - It is difficult to recreate boredom/fatigue of a real flight (1) - Available simulated scenarios dependant on software packages (1) - Pilot may “freeze” when faced with a real flight/actual passengers (1)	8

9.10 9626/12 Feb/March 2025 · Question 11

Question	Answer	Marks
11	EIGHT marks available. Benefits (max 7): e.g. - Can provide results that are generally difficult to measure (1st) - such as e.g. reaction times. (1) - Can simulate a range of scenarios (1) - Some scenarios do not happen frequently in real life (1) - Can simulate driving in different vehicles. (1) - Reduces anxiety level//improves confidence of (new) learners. (1) - Can be repeated any number of times. (1) - Can be customised/personalised to the learner’s needs. (1) - Simulation can be recorded (1) - Can analyse mistakes (1) and suggest solutions (1) - Simulation can be frozen to allow for questions/discussion (1)	8

Drawbacks (max 7)

e.g.

- May not have enough data/information to create some scenarios. (1)
- May not accurately predict the occurrence/effects of human behaviour. (1)
- Might provide misleading scenarios if information was incorrectly/erroneously inputted/added. (1)
- Can require high-end hardware for optimal performance increasing price. (1)
- May give drivers a false sense of security regarding their ability to drive. (1)
- Not a real-world experience (1)

10 Database and file concepts

10.1 9626/12 May/June 2024 · Question 1

Question	Answer	Marks
1(a)	Two from: - It is a primary key in/from another table (1st) - A field in CUSTOMERS table (1) - OR - a primary key that is referenced/used in another table (1st) - A field in CUSTOMERS table (1) - Is used to link/create relationships between tables (1) (may be awarded where candidate has already identified that different tables exist)	2
1(b)(i)	One mark available: - Referential integrity forces table relationships to be consistent (1) Enforces table relationships (1) - Ensures that both sides of the relationship exist (1) - Ensures that any foreign key references an existing primary key (1)	1
1(b)(ii)	Three from: - It avoids/eliminates redundant/unnecessary data (1) - It avoids orphan data (1) (Ensures that) every foreign key value has a matching value in the corresponding primary key// Referential integrity can be used to ensure foreign keys are valid (1) - Referential integrity prevents//warns of the deletion of related records (1) - Value/data within the related fields cannot be changed (1)	3
1(c)	Five from: - A relational database consists of more than one table (1st) - Each table holds data/information about one area/that is related (1) - Tables are linked to each other//relationships are established between tables...(1st) - using primary key (fields) (1) foreign key (fields) (1) compound key (fields) (1) - Using one to one//one to many relationships (1) - Can create queries using more than one table (1) - Can create reports using more than one table (1)	5

10.2 9626/11 Oct/Nov 2024 · Question 4

Question	Answer	Marks
4(a)(i)	ONE mark: Each record in one table relates to only one record in another table (1).	1
4(a)(ii)	ONE from:	1

	Each record in one table can relate to many records in another table (1)	
4(b)	THREE from: - Need to create a link table (1) - Is created using two one-to-many relationships (1) Each primary key must exist as a foreign key (1st) - in the link table (1) (Only award once) - The primary key(s) link(s) to the foreign key(s) (1st) - in the link table (1) (only award once) - Not possible to create a single many-to-many relationship (1) - Only a theoretical type of relationship//cannot exist (1).	3

Question	Answer	Marks	Guidance
9(a)	ONE mark available: - A database that stores data in more than one linked table (1) - It has linked tables (1)	1	Do Not award - Entities - Multiple databases - Multiple locations - Answers such as ‘two tables with relationships’ Use of relationship here – mark as REP
9(b)(i)	TWO from: - Contains unique data (1) - Cannot contain a null value//must contain a value (1) - Can be based on one cell OR composite/based on two cells (1) (whole answer required) Only one per table (1)	2	
9(b)(ii)	ONE from: - It enables links to be formed between tables (1) - It uniquely identifies a record (1) - It may be used in a different table as a foreign key (1) - It maintains accuracy/consistency of data (1) - It allows for quick access to individual records (1)	1	

Question	Answer	Marks	Guidance
9(c)	ONE mark available: • A compound key is two or more pieces of data//a primary key (for an individual record) is a single piece of data (1) • A compound key is based on more than one column//a primary key is based on one column (1) • A compound key gives uniqueness from two columns of data (or more)//a primary key gives uniqueness from one column of data (1)	1	
9(d)	FOUR from: • All data must be atomic (1) • There must be a primary key (1) • There must be no duplicated rows/columns (1) • There must be only one value for each row of each column (1)	4	
9(e)	FOUR from: • Any change to one record which is needed can instantly be made to any • related records/improved data consistency (1) • Provides referential integrity (1) • The database does not have redundant data (1) • Makes the file size small/less storage needed (1) • There is no/less data duplication/repeating data (1st) so there are fewer errors in the data (1) so there is less chance of storing incorrect copies of the data (1) • There is less data to modify (1) • Change (to data) can be made more quickly (1)	4	Do Not Award • No repeat fields

10.4 9626/12 May/June 2023 · Question 8

Question	Answer	Marks
8(a)	Four from: - One mark for identification of any three forms (1) - A method of organising data tables (1) - Attempts to avoid data loss (1) - Attempts to avoid data redundancy//data duplication (1) - Attempts to improve/improves data base efficiency (1) - Breaks data from non-atomic structure to atomic//description of atomisation (takes an example of non-atomic data and states how it becomes atomic) (1) - Breaks data into smaller/other units/tables of data (1) - The units/tables are related to each other (may exemplify) (1) - Only related data is stored in a table (1) - It is an iterative process// It is a multi-step/stage process (1) ... (where) each step/stage improves the efficiency of the database (may be given as a standalone mark) (1)	4
8(b)	Six from: - Searching a large database is quicker to locate records (1) - A relational database does not have duplicated/redundant data (1) - Uses reduced storage space (DNA 'memory') (1) - When data in one field of a table is modified the data in the corresponding field//related field is updated (1) - Integrity constraints ensure that table relationships are/remain valid (1) - It is easier to maintain security of data (1) - Use of related tables reduces need for (as much) data entry (1) Reduced chance of data entry errors (1) Easier for users to produce reports (1)	6

10.5 9626/12 Feb/March 2024 · Question 2

Question	Answer	Marks
2	Eight available. Max two for each matched pair:	8

	• Table name/title (1st) ...a unique/meaningful name for each table (in the database)//to give context (1) • Field name/title (1st) ...a unique/meaningful name to identify each field//to give context (1) (Candidate may describe a meaningful name by example – e.g. ‘full name to hold first and surname’) • Data type (1st) ...Such as (e.g.) integer/alphanumeric • Field length/size (1st) ...specify the number of characters (in each field) (1) • Field default value (1st) ...the value automatically entered when a new record is created (1) • Description of each field (1st) ...explanation of the content (of each field) (1) • Required/null field (1st) ...whether a field can be left blank (1) • Validation (1st) ...test to be applied to data that is entered (1) ...to ensure data entered is sensible/reasonable (candidates may describe this instead of expressly stating it) (1) • Table security (1st) ...who has access to edit (accept examples, such as write or delete) the table • Primary Keys/Foreign keys (1st) ...to identify unique records (1) ...to link tables (1) • Relationships (1st) ...types of relationship between tables are stated (1)	
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10.6 9626/11 May/June 2024 · Question 9

Question	Answer	Marks
9	Max six if concentrates on static or dynamic only Positive Max 6 from: • Static parameter query is fixed/doesn't change/always uses the same criteria (1st) – Which is set when the query is (first) written (1) – saves time when running the query (1) – less likely to have an error because no user input required (1) • A dynamic parameter query can be used to search for different values (1st) – each time it is run (1) – which makes it more flexible (search for other cities) (1) – uses a dialogue box for the user to enter the data required (1) • For dynamic query saves time in designing the query as you don't need to create similar queries (1) • Static queries are more user friendly (1st) – Because it's attached to a button/can click on a button to make it run/don't have to enter search criteria (1)	8

	Negative Max 6 from: - In a static query every time that the query is run it will search for the same data (1st) - cannot be changed/search for another city (1) - can end up having a lot of different static queries (1) - making it difficult to find the query you want (1) - If different data is to be searched for the user would need to open up the query in design view (1st) - and change the data in the criteria to that required (1) - Dynamic parameter query requires less technical knowledge of the user. (1) - Static queries require increased storage space/file size (allow converse, but award once only) (1) Two from: - Statement of when static parameter queries are the better option (1) - Statement of when dynamic parameter queries are the better option (1)	
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10.7 9626/11 May/June 2025 · Question 10

Question	Answer	Marks
10(a)	One mark per bullet point to a maximum of two marks. - Are used to summarise the contents of a table//summarise field values (1) - Also called Group-By queries/aggregate queries (1) - use aggregate functions (1) - They are used to extract aggregate of data items for a group of records - Uses functions such as//calculates (TWO of) SUM, AV(ERA)G(E), MIN, MAX, COUNT	2
10(b)	One mark per bullet point to a maximum of two marks. - Searches using more than one parameter value (1) - Searches on more than one field (1) - It often uses Boolean operands (1) - It is made up of (TWO of) AND, OR or NOT operands or a mixture of these (1)	2
10(c)	One mark per bullet point to a maximum of two marks. - A nested query is a query within another query (1) - It can be often referred to as a subquery (1) - Use the result of one query (1st) - as an input parameter of another (1) - Innermost subquery is executed first, then next level, until the main query is reached. (1)	2

10.8 9626/12 Oct/Nov 2023 · Question 8

Question	Answer	Marks
8	Eight from: - Files created in one software package are sometimes difficult to use/open in other software (1st) - particularly if the packages are incompatible (1) - Generic file formats are compatible with different software packages (1) - When transferring a file to a different computer/device it might not load properly (1st) - since generic file formats are needed to enable different operating systems to read the file/because the particular file format will not operate on that OS (1) - Two marks plus extensions for two examples of the use of generic file types (for a total of 4 marks). e.g. - Word processing files should be saved as a .txt file type (1st) - so that database software can understand it (1) - Database files should be saved/exported as a .csv/.txt/.rtf file type (1st) - so that word processing software can understand it (1) Max 6 marks for just bullet points	8

10.9 9626/11 May/June 2024 · Question 4

Question	Answer	Marks
4	Eight from: - Allows records to be accessed in the order they were entered (1) - Allows records to be accessed randomly (1) - Each record is given index (1) - A database may have several indexes (1st) – based on the information required (1) - Each record is given index (1) - It is a method of allowing fast retrieval (1) - Data is organised into records which consist of fixed length fields (1) - Uses a set of hash tables (1st) – contain "pointers" into the records (1) - Individual records can be retrieved without having to search the entire file (1) - Can be searched quickly (1) - Searches use an index (1st) – which will narrow down the records/data to be searched (1) – that section of the file is then searched sequentially to find the record required. (1)	8

Question	Answer	Marks
3	SIX from: • MIS is used in decision making (1) • MIS is used to coordinate/control data in a business. (1) • MIS software can provide past, present and predictive (TWO FROM) information (1) • MIS software can be used with project management (1) • MIS system may be based on a database (1st) • which can be queried (1) • Can be used to gather/analyse/assess information (1st) • e.g. personnel/sales/inventory/production/employee performance (1) • MIS software could be used to create reports. (1) • By comparing reports to previous reports, trends can be identified (1) • Can produce graphs/charts (1)	6

11 Video and audio editing

11.1 9626/12 Oct/Nov 2023 · Question 10

Question	Answer	Marks
10(a)	Two marks for an explained reason: e.g. - To remove unwanted/unnecessary parts at start/end (1) to make it less boring (1) - To shorten the length (1) because it is too long//to reduce file size (1) - To bring video and audio into synch (1) so that video looks more natural (1) - To make it more exciting (1) by removing boring introduction (1)	2

Question	Answer	Marks
10(b)	Two marks for an explained reason: e.g. - What has been said is shown on the screen//can be read (1) so that nothing is missed (1) - Viewers of a video may have hearing difficulties (1) so subtitles help them understand what is being said (1) - There may be a lot of background noise where the video is being watched (1) so that nothing is missed (1) - Video may be being viewed in a quiet area (1) where having the audio on would bother other people (1) - People watching the video might speak a different language//accent//be muttering (1) so subtitles in other languages can be provided (accept subtitles could be presented in the same language, so words can be read) (1)	2
10(c)	Two marks for an explained reason: e.g. - It can invoke an emotional response from the viewer (1) to increase engagement/enjoyment (1) - To grab audience attention (1) by making the video more interesting (1) - A narrator can provide further content (1) to drive home the emotion/mood//can clarify a video (1) - To add sound effect (1) can heighten the impact of (e.g.) explosions/speeding cars (1) to add authenticity/context/reality (1) - To add a clear soundtrack (1) without background noise (1)	2

11.2 9626/13 May/June 2024 · Question 8

Question	Answer	Marks
8(a)	Two marks available for a full explanation: To identify the people involved in the creation of the video (1) so they can get recognition//acknowledge their role in creating it (1)	2

8(b)	Two from: • A frame from the video to be used as a photograph/thumbnail (1) used for (e.g.) marketing/publicity//to check for continuity//to be used as an icon (1)	2
8(c)	Two from: • Not all devices/platforms are able to play videos in all formats/the original format (1) • which means some videos are incompatible with specific devices/platforms • so this makes the video able to play/be accessed/used on a wider range of devices/platforms (1) • Videos may need to be converted to reduce their file size (1st) • to allow them to be attached to an email (1) • reduce storage space (1) • upload to video streaming website/social media (1)	2

Question	Answer	Marks	Guidance
5(a)	ONE mark available: - It has a width of 16 units and height of 9 units (1) width to height ratio is 16:9/1.78:1 (1)	1	Do Not Award Length not allowed. Unless it is length and height
5(b)	TWO marks available for two complete explanations: - Ensures high-definition quality therefore final product will be of a high standard/clear to see (1) - Offers more content to the viewer so a better-quality product (1) - Is the more modern format, so product looks up-to-date (1) - Is the most widely used format in computer monitors/Television so the video will fit properly (1) 16:9 is the international/industry standard therefore it will be more widely used/meets international broadcast standards (1)	2	

11.4 9626/12 Oct/Nov 2025 · Question 8

Question	Answer	Marks
8(a)	One from: - Compressed data cannot be reconstructed (1) - It permanently removes data (1)	1
8(b)	One from: - Compressed data can be reconstructed (1) - Does not permanently remove data (1)	1

Question	Answer	Marks
8(c)	Four available. One per instance of a valid comparison: e.g. - Lossless will not affect the quality of video/lossy will affect the quality (1) - Lossy reduces the colour range (1) - Lossy can alter the frame rate (make the video jumpy/stutter) (1) - Lossy can desync the audio with video (1) - Lossy can affect the colour settings of the video (1) - Lossy can affect resolution of video (1) - Lossy can affect the bitrate/make incompatible with devices (1)	4

11.5 9626/12 May/June 2024 · Question 8

Question	Answer	Marks
8(a)	Two from: - To remove unwanted parts of the clip//there are unwanted clips//parts (accept example) (1) - To reduce the length/time of the track//clip is too long (1) - To reduce file size//file size is too big//storage space required (1) - To make same length as an accompanying video clip//other example of fitting to an event (1)	2
8(b)	One from: - To make two (or more) audio clips flow into each other (1) - To combine the best parts of one clip with the best parts of another clip/combine the best parts of several recordings made from the same concert/act (1) - To reorder audio clips (1) - To give continuity/continuation//no gaps between them//sound files (1) - To create a single track/clip (1) - To make the file longer//a suitable length (1)	1
8(c)	Three from: - To boost/adjust the frequencies (in a sound signal) (1st) - By raising (1) or lowering (1) - Can use filters (1)	3

11.6 9626/12 Feb/March 2025 · Question 10

Question	Answer	Marks
10(a)	One mark per bullet point to a maximum of two marks. - Sets the peak amplitude/loudest part of a track to a specific level. (1) - Makes multiple tracks have the same gain/peak/relative volume. (1) - Helps increase the volume of an entire audio track proportionally. (1) - Helps achieve consistent volume levels across different audio files. (1) - Helps maximising overall volume levels without compromising sound quality (1)	2
10(b)	One mark per bullet point to a maximum of two marks. - Eliminates/removes unwanted sounds. (1) - Makes audio track/signal clearer. (1) - Plays a mirror image of the original audio track (1st) - to partially cancel noise out. (1)	2

11.7 9626/11 and 9626/13 Oct/Nov 2025 · Question 8

Question	Answer	Marks
8(a)	Max one for: - It means adding tracks/audio clips (1) Max one mark for any valid reason: e.g. - Add complexity to the music/audio (1) - Adds voiceover (1) - Add extra instruments (1) - Add sound effects to match what is happening on the screen (1)	2
8(b)	Max one for: - It means boosting/reducing/changing the levels of different frequency bands (1) Max one for any valid reason: e.g. - To pick out speech from background sounds (1) - To reduce the hiss (1) - To reduce/increase identified/intrusive noise frequencies (1) - To emphasise/highlight a guitar solo in a music track (1)	2
8(c)	Max one for: - Is an effect that adds echo to a sound (may describe sound being bounced off (e.g.) walls) (1) Max one for any valid reason: e.g. - To simulate being in a different environment (e.g. cavern/concert hall) (1) - To give depth/richness to a sound (1) - To create a dramatic effect in a radio/audio play/podcast (1)	2

11.8 9626/13 Oct/Nov 2023 · Question 7

Question	Answer	Marks
7(a)	Two from: - Fade in occurs at the beginning of a sound clip when the sound gradually gets louder (1) - Fade out occurs towards/at the end of a sound clip when the sound gradually gets softer (1) - Volume stops increasing when the required sound level is reached (1) - Sound disappears altogether (1)	2
7(b)	Two from: - Lossless compression retains the quality of the original/does not reduce the quality of the audio file (1) - Data is not lost (1) - When lossless compression software sees a repeated sequence of bits it replaces the repeated sequences with a character (1st) - which indicates what is being repeated and how many times (1)	2
7(c)	Two from: - Removes some of the audio file's data (1) - Reduces the quality of the audio file (1) - Removes sounds at very low and high frequencies (which are outside the range of human hearing) (1) - Creates smaller file sizes than lossless compression (1) - Once this data has been removed from the files it can never be replaced (1)	2

11.9 9626/12 Feb/March 2024 · Question 10

10(a)	Two available: - The number of samples (1st) - within a given period of time (accept ANY time unit) (1) - usually within 1 second (1).	2
10(b)	Max two for one complete explanation: - The sampling rate can affect the quality of audio files/how the audio sounds (1st) - because it captures sounds at discrete points (and not the whole sound) (1) - A higher sampling rate sounds better than a lower rate (1st) - because a higher sampling rate more accurately represents the original sound (1) - The lower the sample rate, the smaller the file size (1st) - because there is less data stored (1)	2

Question	Answer	Marks
10	Five from: • Sampling resolution is the number of bits (1st) – per sound sample (1) • The higher the sampling resolution (1st) – the more accurately the (final) outcome (of the process) will represent the sound made (accept converse argument for whole answer) (1) • The higher the sampling resolution/the greater the resolution the greater the size of the file (accept converse) (1) • Modern day digital audio is normally found in 16 bit/24 bit resolutions • 8-bit resolution can take one of 256 different values (1st) – which is not generally considered enough resolution to accurately represent music audio (1) • 16-bit resolution can have one of 65 536 different values (1) • 24-bit stores 16 777 216 different values (1) • the difference between 24 and 32- bit cannot be discerned by human hearing (1) • These values are all per channel (1)	5